

CMPUT 654: Mathematical Foundations of Modern AI Systems

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Lecture 4: Sequence modeling, generation, and memory

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About these lecture notes. We know how to fit a sequence model by next-token prediction. We now ask whether fitting successful examples makes the model generate successful sequences. The answer depends on how prediction errors affect later generation, what cost we care about, and what structure the model preserves. This leads from log loss to error recovery, reverse KL, and models with explicit memory.

Scope relative to class. Further questions and ideas arose while preparing these notes, so they extend and refine the discussion from the lecture. The lecture developed the positive-data fitting question, the binary error-propagation example, stability of Markov models, and the concentration-dependent reverse-KL bound. It then introduced state-space models, latent likelihoods, the discrete-to-vector representation, and the system-identification approach of regression followed by factorization. The meeting ended before Cayley–Hamilton and the detailed LDS results. The notes group the cost bounds first, then use exact fixed-KL cost intervals to develop two laws attaining the cost bounds. We reuse these laws when interpreting normalized loss, comparing fitting objectives, and introducing recovery. Repeated errors, parity, and the explicit modulus calculations are developed in exercises. A further section asks how parameter sharing affects learning as sequence length grows. The fitting and structural choices follow. Headings ending in a superscript asterisk* mark supplementary material: sharper comparisons between likelihood and cost, a finite-sample fitting guarantee, and the LDS results. The exercises retain the calculations and proofs left to students. Longer proofs, a statistical obstruction to reverse-KL fitting, and the variational discussion are in the appendices.

Notation and reading guide. Tokens are $X_t \in [N]$; $Z_t = e_{X_t}$ denotes their vector encoding. Later, Z_t also allows general real-vector observations. Hidden states are S_t . A vector $x \in \mathbb{R}^d$ is a column vector. For indices $a \leq b$, write $X_{a:b} = (X_a, \dots, X_b)$ and $X_{<t} = X_{1:t-1}$; lowercase $x_{a:b}$ denotes a realized sequence. For a symbol u , $u^k = (u, \dots, u)$ is the length- k constant sequence, so $0^T = (0, \dots, 0)$, and juxtaposition denotes concatenation, as in $10^{T-1} = (1, 0, \dots, 0)$. Sequence notation is useful because the same object has two roles: it is an ordered tuple in $[N]^T$, and it is a time-indexed function on $[T]$. We use tuple notation when order and prefixes matter, and ordinary vector notation when applying linear algebra. For a sequence cost $c : [N]^T \rightarrow \mathbb{R}$ and a distribution P over $[N]^T$, $\mathbb{E}_P[c]$ abbreviates $\mathbb{E}[c(X_{1:T})]$ for $X_{1:T} \sim P$. Thus

$$\mathbb{E}_P[c] = \sum_{x_{1:T}} P(x_{1:T}) c(x_{1:T}).$$

By default, in what follows, P is the data law and Q , or Q_θ , is a model law. We use κ for a KL value or budget, s for a cost-discrepancy value or budget, and η for per-token accuracy, including the optimization tolerance introduced in [Section 4.4.3](#). Logarithms are natural. Read through [Section 4.8](#); the supplementary material can be read separately. Numbered endnotes collect short refinements that may be skipped on a first reading.

4.1 From next-token prediction to useful generation

Lecture 1 described transformers as maps from token prefixes to next-token distributions. Lecture 2 asked what structure allows learning to generalize. Lecture 3 explained why log loss is a natural objective for probability estimation. Pretraining brings these ideas together: shared parameters are fitted to predict tokens across many sequences and contexts. The prediction problem supplies a scalable way to learn from data.

Throughout the discrete discussion, sequences have fixed length T over a finite vocabulary $[N]$, and costs are real-valued. The finite space $[N]^T$ simplifies the arguments: every cost is bounded, and expectations are finite sums. Many results hold more generally; [Appendix 4.A](#) explains where finiteness is used and what can replace it. It is also easy to extend the results to allow variable length sequences as outlines in Lecture 1. For simplicity, we restrict our attention to the fixed length case.

Let $X_{1:T}^{(1)}, \dots, X_{1:T}^{(n)}$ be independent training sequences from P , a distribution over $[N]^T$. Their tokens may have arbitrary dependence within each sequence. Empirical log-loss minimization chooses

$$\hat{\theta} \in \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n -\log Q_{\theta}(X_{1:T}^{(i)}). \quad (4.1)$$

The chain rule of probabilities gives that for every $x_{1:T} \in [N]^T$,

$$Q_{\theta}(x_{1:T}) = \prod_{t=1}^T Q_{\theta}(x_t \mid x_{<t}), \quad x_{<t} = (x_1, \dots, x_{t-1}). \quad (4.2)$$

Every sequence law has this factorization. We call it the *autoregressive factorization*; restrictions on the conditional predictors, such as a transformer architecture or a finite context window, specify the model class. Taking logarithms in [Eq. \(4.2\)](#) and plugging in $X_{1:T}$ in place of $x_{1:T}$ gives

$$-\log Q_{\theta}(X_{1:T}) = \sum_{t=1}^T -\log Q_{\theta}(X_t \mid X_{<t}). \quad (4.3)$$

Consequently, next-token log-loss minimization, the workhorse of language model training, is sequence maximum likelihood.

When evaluating [Eq. \(4.3\)](#), the predictor receives the observed prefix, including the observed next input. This is *teacher forcing*: the student's cost is measured at the teacher's data. The metaphor comes from viewing the problem as a teacher generating data that a student needs to use to learn. During generation, the next input is instead sampled from the model:

$$\tilde{X}_t \sim Q_{\theta}(\cdot \mid \tilde{X}_{<t}). \quad (4.4)$$

The resulting sequence has joint law Q_{θ} . The same factorization supports both likelihood evaluation and sampling.

As explained earlier, the parameter $\hat{\theta}$ from [Eq. \(4.1\)](#) is maximum likelihood parameter estimate: minimizing the negative log of a product maximizes that product, which, under independence, is the probability assigned to the data. For a fresh sequence $X_{1:T}$ with law P , the log-loss identity from Lecture 3 and the decomposition in [Eq. \(4.3\)](#) give

$$\begin{aligned} \mathbb{E}[-\log Q_{\theta}(X_{1:T})] &= H(P) + D(P, Q_{\theta}), \\ D(P, Q_{\theta}) &= \sum_{t=1}^T \mathbb{E}[D(P(\cdot \mid X_{<t}), Q_{\theta}(\cdot \mid X_{<t}))]. \end{aligned} \quad (4.5)$$

The entropy term is independent of θ . Thus the population objective minimizes KL; the empirical objective estimates it from data. The second identity is the *KL chain rule*: sequence KL is the sum of the expected conditional KLs. [Exercise 1](#) asks for its derivation.

In ML/statistics we distinguish between data and model laws. This leads to the following definition:

Definition 4.1.1 (Forward and reverse KL). When P is a data law and Q is a model law, we call $D(P, Q)$ the *forward KL*, while $D(Q, P)$ is called the *reverse KL*.

With our finite spaces, we have

$$\begin{aligned} D(P, Q) &= \sum_x P(x) \log \frac{P(x)}{Q(x)} \quad (\text{the forward KL}), \\ D(Q, P) &= \sum_x Q(x) \log \frac{Q(x)}{P(x)} \quad (\text{the reverse KL}). \end{aligned}$$

Note that above a zero numerator contributes zero; a positive numerator with zero denominator contributes $+\infty$.¹ The names refer to the roles of P, Q : forward KL averages under the data law, whereas reverse KL averages under the generated law.

As explained, $\hat{\theta}$ is expected to approximately minimize $D(P, Q_\theta)$ in θ . The discrepancy $D(P, Q_{\hat{\theta}}) - \inf_\theta D(P, Q_\theta)$ describes the effect of using a finite dataset in the lack of access to P (and a reliable way of minimizing $D(P, Q_\theta)$). Recall from Lecture 2 that this is expected to be controlled when n is large and things are “sufficiently nice”. As such, we leave this question alone in this lecture and instead ask, would it be even beneficial (and to what degree) if we could find a model Q with a small population KL?

4.1.1 A task specifies which generations are useful

A prompt asks the model to continue in a particular way. Mathematically, *prompting* uses a conditional distribution $Q_\theta(\cdot \mid u)$, where u is the supplied prompt (we omit the full formal description of the setup, but here u is also a sequence in the typical language model training setting). For a mathematics task, a useful continuation might be a correct solution; for a knowledge question, an accurate answer. A *cost* $c(u, x_{1:T})$ assigns a numerical value to the generated response. Given a test-prompt distribution ν , the expected cost of a model Q_θ is

$$\mathcal{J}(\theta) = \mathbb{E}[c(U, \tilde{X}_{1:T})], \quad U \sim \nu, \quad \tilde{X}_{1:T} \mid U \sim Q_\theta(\cdot \mid U). \quad (4.6)$$

A model’s quality then is determined by $\mathcal{J}(\theta)$. The cost function is thus $c : [N]^T \rightarrow \mathbb{R}$. In what follows, without loss of generality, we assume that it is nonnegative valued. Sometimes (especially in RL) the task is formulated with rewards. These formulations are equivalent: Maximizing expected reward is equivalent to minimizing its negative.

A common approach to get a small-cost model is to collect low-cost (or even zero-cost) prompt–response pairs, and then find a model that minimizes the empirical log-loss, aiming to “mimick” the responses in the training data. Formally, solve

$$\hat{\theta} \in \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \sum_{t=1}^T -\log Q_\theta(X_t^{(i)} \mid U^{(i)}, X_{<t}^{(i)}). \quad (4.7)$$

In training large language models, often incremental methods are used, such as some form of gradient descent. Lectures 9–11 return to how gradient-based optimization selects among fitted solutions

and when local gradients lead to global convergence. When this is the case, typically people start gradient descent with a parameter θ_0 which is obtained during *pre-training* (as outlined in Lecture 1). Then, with the data as above, people continue gradient descent from θ_0 with the data as above. This practice is called *fine-tuning* by likelihood on *positive examples* (starting from a pretrained model parameters). The data is called “positive” because it will contain completed proofs (in math problems) or, more generally, answered deemed correct, with no examples in the data of incorrect responses.

For simplicity, in this lecture we do not consider the “continuing” aspect of the fine-tuning and will ask the question whether minimizing log-loss with positive data will produce low-cost answers to new prompts. To isolate the connection between fitting and task performance, we set the prompts aside; keeping the prompts (conditioning) around would introduce extra notation, but otherwise, in general, changes nothing. There is one exception to this which will be noted when it arises.

With this, let us turn to formulating the specific question studied next. Let P be the law of positive sequences, Q the fitted law, and c the cost. One should generally think that $\mathbb{E}_P[c] = 0$; the results will aim at this case but will be more general. Recall from Eq. (4.5) that the population log loss is

$$\mathbb{E}_P[-\log Q] = H(P) + D(P, Q).$$

As explained earlier, one should think about log-loss minimization, or likelihood fitting as seeking a model Q that with small $D(P, Q)$. Yet, the task calls for small generated cost $\mathbb{E}_Q[c]$. Suppose fitting gives us $D(P, Q) \leq \kappa$. The basic question we ask is how large can $\mathbb{E}_Q[c]$ be? In other words, is controlling KL this way sufficient to control generation cost? To what degree? What are the strengths and weaknesses of this approach? Naturally, we will continue with asking whether there is anything better.

4.2 How does KL control the task cost?

Define the *cost discrepancy*

$$d_c(P, Q) = |\mathbb{E}_P[c] - \mathbb{E}_Q[c]|$$

to measure the change in expected cost from the data law to the model law. For cost-free data laws, thank to the assumption that $c \geq 0$, $d_c(P, Q) = \mathbb{E}_Q[c]$, so a bound on this discrepancy answers our question.

We will compare four ways to turn KL control into cost control, starting with simpler and moving to more elaborate results. We will compare the resulting bounds, and ask which of them are sharp and when. A supplementary subsection asks for both the smallest and largest cost discrepancies at a fixed KL, relating these to moduli of continuity. We then compute these intervals in binary examples to see how differently models with the same likelihood can behave.

4.2.1 Use the cost range

Recall the definition of the total variation distance between probability laws: In what follows we fix a finite set $\mathcal{X}$ as the domain of the cost and the probability distributions P and Q .

Definition 4.2.1 (Total variation). For laws P, Q over a finite set $\mathcal{X}$, their *total-variation distance* is

$$\|P - Q\|_{\text{TV}} = \sup_{A \subseteq \mathcal{X}} |P(A) - Q(A)| = \frac{1}{2} \sum_{x \in \mathcal{X}} |P(x) - Q(x)|. \quad (4.8)$$

Lecture 3 explains why the same formula is a norm on signed measures and a metric on probability laws, and contrasts both with a pseudometric.

Proposition 4.2.2 (Total variation controls bounded costs). *If $0 \leq c(x) \leq C_{\max}$ for every $x \in \mathcal{X}$, then*

$$|\mathbb{E}_Q[c] - \mathbb{E}_P[c]| \leq C_{\max} \|P - Q\|_{\text{TV}}. \quad (4.9)$$

This is Lecture 3, [Exercise 4\(c\)](#), after scaling the cost by $C_{\max}$. The exercise proves it from the positive and negative parts of the signed measure $Q - P$.

This bound is exact when maximizing over costs with the specified range: $c(x) = C_{\max} \mathbb{I}\{Q(x) > P(x)\}$ attains equality. Thus TV measures the largest expected-cost discrepancy among such costs.

Pinsker's inequality from Lecture 3 now gives

$$\mathbb{E}_Q[c] \leq \mathbb{E}_P[c] + C_{\max} \sqrt{\frac{D(P, Q)}{2}}. \quad (4.10)$$

For any bounded cost, sufficiently small KL therefore gives sufficiently small cost discrepancy. If $\mathbb{E}_P[c] = 0$, a sufficient condition for $\mathbb{E}_Q[c] \leq s$ is

$$D(P, Q) \leq 2(s/C_{\max})^2.$$

Both the square-root dependence and its constant are necessary for general bounded costs.

Example 4.2.3 (A locally sharp fair-coin comparison). Let $\mathcal{X} = \{0, 1\}$,

$$P = \text{Bernoulli}(1/2), \quad Q = \text{Bernoulli}(1/2 + \delta), \quad c(x) = x.$$

As $\delta \rightarrow 0$,

$$d_c(P, Q) = |\delta|, \quad D(P, Q) = 2\delta^2 + O(\delta^4).$$

Pinsker's bound divided by the actual discrepancy tends to one: it is *locally sharp*.

For a count of token errors, $C_{\max} = T$, so the horizon also enters the conversion.

Recall that we are interested in the case when $\mathbb{E}_P[c] = 0$ or small, while $c \geq 0$. Can we improve this bound using this information?

4.2.2 Successful data give a sharper bound*

Let us start with the case when $\mathbb{E}_P[c] = 0$.

Then, the following result applies:

Proposition 4.2.4 (Forward KL and zero-cost data). *If $0 \leq c \leq C_{\max}$, $\mathbb{E}_P[c] = 0$ then*

$$\mathbb{E}_Q[c] \leq C_{\max}(1 - e^{-D(P, Q)}) \leq C_{\max} D(P, Q). \quad (4.11)$$

This is an improvement over [Eq. \(4.10\)](#) when $D(P, Q) \leq 1/2$ and the gap between them grows as the KL between P and Q approaches zero. For small KL, exploiting successful data replaces the square-root dependence from Pinsker by linear dependence.

Recording only whether an event occurred can only decrease KL. For an event certain under P , this gives a lower bound on its probability under Q . Recall that for $p, q \in [0, 1]$, $\text{kl}(p, q)$ denotes the KL between the Bernoulli distributions with parameters p and q .

Proposition 4.2.5 (From distribution KL to event probability KL). *Let P, Q be distributions over the finite domain $\mathcal{X}$. Then, for any $A \subset \mathcal{X}$,*

$$\text{kl}(P(A), Q(A)) \leq D(P, Q).$$

In particular, if $P(A) = 1$ then

$$Q(A) \geq e^{-D(P, Q)}.$$

Proof. Consider the transformation that maps any $x \in \mathcal{X}$ to 1 if $x \in A$ and to 0 otherwise (this is just the indicator if A). Apply the data processing inequality with this deterministic transformation to get the first inequality. Indeed, the push-forward of P (Q) under $\mathbb{I}_A$ is the Bernoulli distribution with parameter $P(A)$ (respectively, $Q(A)$). To get the second inequality, note that if $P(A) = 1$, $\text{kl}(P(A), Q(A)) = \log \frac{P(A)}{Q(A)}$. Rearrangement then gives the second inequality. The general data-processing statement and its log-sum proof appear in Lecture 3, [Proposition 3.4.2](#) and [Exercise 4](#). $\square$

With this we can return to the proof of [Proposition 4.2.4](#).

Proof of Proposition 4.2.4. Let $A = \{x : c(x) = 0\}$. Note that since $c \geq 0$ and $\mathbb{E}_P[c] = 0$, $P(c = 0) = P(A) = 1$. Now, by algebra, using the result of the previous proposition and since $1 - e^{-u} \leq u$ for $u \geq 0$,

$$\mathbb{E}_Q c \leq C_{\max} Q(A^c) = C_{\max}(1 - Q(A)) \leq C_{\max}(1 - e^{-D(P, Q)}) \leq C_{\max} D(P, Q). \quad \square$$

The bound is sharp: Let c take on values of 0 and $C_{\max}$, but otherwise be arbitrary. Let P be as in the proposition. Let R be any probability distribution supported on $\{c > 0\} = \{c = C_{\max}\}$. For any $0 < a < 1$, choosing $Q = (1 - a)P + aR$, we have $D(P, Q) = -\log(1 - a)$ and $\mathbb{E}_Q[c] = aC_{\max}$. Hence, indeed

$$\mathbb{E}_Q[c] = aC_{\max} = C_{\max}(1 - e^{-D(P, Q)}).$$

Thus, equality holds in the first bound of [Eq. \(4.11\)](#).

4.2.3 Concentration and the direction of KL

Our next result strengthens the bound we obtained using Pinsker inequality when the costs under P concentrate around their mean, but the price is switching from forward to reverse KL:

Proposition 4.2.6 (Cost control from exponential moments). *Let P, Q be distributions over a finite set $\mathcal{X}$ and let $c : \mathcal{X} \rightarrow \mathbb{R}$. Then.*

$$\mathbb{E}_Q[c] - \mathbb{E}_P[c] \leq \inf_{\lambda > 0} \frac{D(Q, P) + \log \mathbb{E}_P[e^{\lambda(c - \mathbb{E}_P[c])}]}{\lambda}. \quad (4.12)$$

This reduces cost control to bounding an exponential moment and optimizing one positive scalar. The exponential-tilting proof is in [Appendix 4.B](#). A quadratic bound on that logarithmic moment function gives a particularly simple answer.

Definition 4.2.7 (Subgaussian cost). The cost c is v^2 -subgaussian under P , for $v \geq 0$, if

$$\log \mathbb{E}_P \left[e^{\lambda(c - \mathbb{E}_P[c])} \right] \leq \frac{\lambda^2 v^2}{2} \quad \text{for every } \lambda \in \mathbb{R}. \quad (4.13)$$

Here v measures a scale of fluctuations; v^2 is the variance proxy. We write $v_P(c)$ for the *optimal variance proxy*, which is also known as the subgaussian constant of c under P :

$$v_P(c) = \min \left\{ a \geq 0 : \log \mathbb{E}_P[e^{\lambda(c - \mathbb{E}_P[c])}] \leq \frac{\lambda^2 a}{2} \text{ for every } \lambda \in \mathbb{R} \right\}.$$

The minimum is attained; a constant cost has $v_P(c) = 0$.²

Theorem 4.2.8 (Reverse KL and cost concentration). *Suppose c satisfies Eq. (4.13). Then, interpreting $0 \cdot \infty = \infty$, we have*

$$|\mathbb{E}_Q[c] - \mathbb{E}_P[c]| \leq v \sqrt{2D(Q, P)}. \quad (4.14)$$

Proof. Substituting Eq. (4.13) into Eq. (4.12) gives

$$\inf_{\lambda > 0} \left(\frac{D(Q, P)}{\lambda} + \frac{\lambda v^2}{2} \right) = v \sqrt{2D(Q, P)}.$$

Applying the same argument to $-c$ bounds the other direction. $\square$

Interchanging P, Q and using the optimal proxies $v_P(c), v_Q(c)$ gives

$$d_c(P, Q) \leq \sqrt{2 \min\{v_Q(c)D(P, Q), v_P(c)D(Q, P)\}}. \quad (4.15)$$

While the forward KL term refines the bound coming from Pinsker's inequality (i.e., Eq. (4.10)), when besides that P is supported on zero-cost sequences no cost information is available while fitting Q , the usefulness of this refinement is limited: $v_Q(c)$ may be as large as $C_{\max}^2/4$ and remains uncontrolled.

The reverse KL part of the bound is more promising in that $v_P(c)$ is something we expect to be small. In particular, when $P(c = 0) = 1$, as noted beforehand, $v_P(c) = 0$. However, this immediately reveals a limitation: The bound tells us that for such P and c , the costs under P and Q either match perfectly (when $D(Q, P) < \infty$) or if the costs are different then $D(Q, P) = \infty$. This foreshadows the difficulty of controlling reverse KL, which we will discuss later.

4.2.4 Use the variability of the data's cost*

Can we use cost variation under P while keeping forward KL, the quantity likelihood fitting controls? As it turns out, the answer is yes. The proof of this is slightly more complicated than the previous ones. The proof centers the cost under P , expands $Q - P$ in powers of $\sqrt{Q} - \sqrt{P}$, and uses Cauchy-Schwarz to obtain a variance term instead of an exponential-moment bound. Write $\sigma_P(c)^2 = \mathbb{E}_P[(c - \mathbb{E}_P[c])^2]$ for the variance of the cost.

Proposition 4.2.9 (Cost variance and forward KL). *If $0 \leq c \leq C_{\max}$ and $D(P, Q) < \infty$, then*

$$d_c(P, Q) \leq 2\sigma_P(c)\sqrt{D(P, Q)} + C_{\max}D(P, Q). \quad (4.16)$$

Appendix 4.C.1 gives the proof.

The combination of a variance-dependent square-root term and a range-dependent linear term is called *Bernstein form*. The square-root term uses the data's cost variation rather than its full range. When $\sigma_P(c) = 0$, this term vanishes, leaving a bound linear in KL. Note that this bound extends the zero-cost data bound Eq. (4.11). Indeed, when $c \equiv 0$, $\sigma_P(c) = 0$ and we get back Eq. (4.11). The

additive term accounts for changes in cost on outcomes to which Q assigns probability, including outcomes absent under P .

The variance-sensitive and subgaussian bounds are useful in different regimes. In [Example 4.2.3](#), this bound is asymptotic to $\sqrt{2}|\delta|$, while Pinsker and the subgaussian bound are asymptotic to the actual discrepancy $|\delta|$. For rare errors, the variance-sensitive bound can instead give the correct order while the subgaussian bound is much larger; [Appendix 4.C.2](#) works out both comparisons.

4.2.5 The play of cost when forward KL is controlled

A sharp upper bound tells us how large the cost discrepancy can be. To understand how much a KL value tells us about task performance, we need both the smallest and the largest cost discrepancies compatible with that value. The binary examples below will give exact intervals: at the same KL, some generators have much smaller cost than others. Calculating these fixed-KL intervals reveals the ambiguity that remains even after the upper bounds have been made sharp. If the interval is large, we can conclude that KL on its own does allow a “large play”. *Knowing how large the play is exactly what allows us to think about whether it is worthwhile to go beyond controlling KL.* The rest of this section develops these ideas and connects them to standard concepts in mathematics.

Fix the data law P and cost c , and let Q range over all probability laws on $\mathcal{X}$. Define the two endpoints

$$\begin{aligned} f_P(\kappa) &= \min_{Q: D(P,Q)=\kappa} d_c(P, Q), \\ g_P(\kappa) &= \max_{Q: D(P,Q)=\kappa} d_c(P, Q). \end{aligned} \tag{4.17}$$

Here KL levels range over the attainable values in $[0, +\infty]$. The extrema exist in our finite setting; [Appendix 4.A](#) gives the compactness argument. These are the sharp bounds in

$$f_P(D(P, Q)) \leq d_c(P, Q) \leq g_P(D(P, Q)).$$

The preceding inequalities studied the upper endpoint. The lower endpoint answers an equally relevant question: *how small could the cost discrepancy be at this same KL?* A large gap between the endpoints means that likelihood alone leaves substantial uncertainty about task performance. Conversely, a small gap means no such wiggle room, which would mean that controlling KL alone could be sufficient to get a good control on the task cost.

From fixed values to error budgets. Usually we know an upper bound on a discrepancy, rather than its exact value. The corresponding sharp conversion functions are called *moduli of continuity*. At the fixed reference P , define

$$\omega_{D \rightarrow c}(\kappa; P) = \sup_{Q: D(P,Q) \leq \kappa} d_c(P, Q). \tag{4.18}$$

This is the modulus of continuity of the cost discrepancy with respect to KL. It asks: *how large can the cost discrepancy be when KL is at most κ ?* In the other direction, define

$$\omega_{c \rightarrow D}(s; P) = \sup_{Q: d_c(P,Q) \leq s} D(P, Q). \tag{4.19}$$

This is the modulus of continuity of KL with respect to cost discrepancy. It asks: *how large can KL be when the cost discrepancy is at most s ?* The arrows specify which discrepancy controls the other; both formulas use forward KL, $D(P, Q)$.

Proposition 4.2.10 (The sharp conversions from fixed-KL endpoints). *For the fixed P, c ,*

$$\begin{aligned}\omega_{D \rightarrow c}(\kappa; P) &= \sup_{\kappa' \leq \kappa} g_P(\kappa'), \\ \omega_{c \rightarrow D}(s; P) &= \sup\{\kappa : f_P(\kappa) \leq s\}.\end{aligned}\tag{4.20}$$

Both moduli are nondecreasing. They are the smallest nondecreasing functions giving, respectively, the bounds

$$d_c(P, Q) \leq \omega_{D \rightarrow c}(D(P, Q); P), \quad D(P, Q) \leq \omega_{c \rightarrow D}(d_c(P, Q); P).$$

Proof. At KL level κ , the largest cost discrepancy is $g_P(\kappa)$, and some cost discrepancy is at most s exactly when $f_P(\kappa) \leq s$. Taking the indicated suprema proves [Eq. \(4.20\)](#). Increasing either budget enlarges its feasible set, proving monotonicity. If a nondecreasing function g satisfies $d_c(P, Q) \leq g(D(P, Q))$, then every Q with KL at most κ has discrepancy at most $g(\kappa)$. Taking the supremum proves $\omega_{D \rightarrow c}(\kappa; P) \leq g(\kappa)$. The other direction is identical. $\square$

Thus the upper endpoints determine the $D \rightarrow c$ modulus, while the lower endpoints determine the $c \rightarrow D$ modulus. In particular, if g_P is nondecreasing, it already equals $\omega_{D \rightarrow c}$. An increasing lower bound can be read in the opposite direction by inverting it.

Proposition 4.2.11 (Inverting a lower bound). *Suppose $f : [0, +\infty] \rightarrow [0, +\infty]$ is strictly increasing and $f(D(P, Q)) \leq d_c(P, Q)$ for every Q . Then*

$$\omega_{c \rightarrow D}(f(\kappa); P) \leq \kappa.$$

If some Q has $D(P, Q) = \kappa$ and $d_c(P, Q) = f(\kappa)$, equality holds. In particular, when the sharp lower endpoint f_P is strictly increasing, $\omega_{c \rightarrow D}(s; P) = f_P^{-1}(s)$ for s in its range.

Proof. If $d_c(P, Q) \leq f(\kappa)$, then $f(D(P, Q)) \leq f(\kappa)$. Strict increase gives $D(P, Q) \leq \kappa$. A law attaining the lower bound at κ also attains this KL budget, proving equality. $\square$

This explains why we care about the converse modulus: it describes the lower-bound side of the same comparison. A small lower endpoint and a large upper endpoint allow models of very different task quality to have exactly the same KL. The examples below will compute both endpoints and exhibit such models.

Uniform comparisons over data laws. If the reference law is unknown, we may want a conversion valid for every P in a class $\mathcal{P}$. The *uniform moduli* are

$$\omega_{D \rightarrow c}^{\mathcal{P}}(\kappa) = \sup_{P \in \mathcal{P}} \omega_{D \rightarrow c}(\kappa; P), \quad \omega_{c \rightarrow D}^{\mathcal{P}}(s) = \sup_{P \in \mathcal{P}} \omega_{c \rightarrow D}(s; P).$$

Enlarging $\mathcal{P}$ can only increase these worst-case conversion bounds. For example, with $0 \leq c \leq C_{\max}$, take $\mathcal{P}_{\sigma} = \{P : \sigma_P(c) \leq \sigma\}$. [Proposition 4.2.9](#) gives

$$\omega_{D \rightarrow c}^{\mathcal{P}_{\sigma}}(\kappa) \leq 2\sigma\sqrt{\kappa} + C_{\max}\kappa.$$

This reformulates the earlier sharpness question: among all data laws whose cost variance is at most σ^2 , how much can cost change under a KL budget κ ? Throughout, Q still ranges over all laws.

The norm-comparison analogy. For two norms on $\mathbb{R}^d$, sharp bounds $\alpha\|z\|_a \leq \|z\|_b \leq \beta\|z\|_a$ place $\|z\|_b$ between αr and βr when $\|z\|_a = r$. The ratio β/α measures the multiplicative spread at this fixed input norm. For norms the sharp budget conversion is linear in the budget r : substituting $z = ru$ gives

$$\sup_{\|z\|_a \leq r} \|z\|_b = r \sup_{\|u\|_a \leq 1} \|u\|_b \quad (r > 0).$$

Norms scale linearly under multiplication, which forces this form. Our cost–KL conversions instead involve functions such as $\sqrt{\kappa}$ and $1 - e^{-\kappa}$. The shared question is how tightly one measurement determines another. Equivalence of finite-dimensional norms also gives a positive lower endpoint when the fixed input norm is positive. The examples below show that the cost discrepancy can instead be zero at positive KL. [Appendix 4.C.3](#) explains the terminology’s connection to continuity and approximation.

We can ask the same questions for other probability metrics, such as TV: which cost discrepancies are compatible with a fixed distance, and what are the sharp conversion functions? The next section will examine TV alongside KL.

4.3 Examples of task cost control at fixed KL

We now use point-mass data distributions to illustrate how tightly fixing forward KL controls task cost. Upper bounds alone leave open whether the smallest and largest compatible costs are close; here we characterize the entire interval and show that they need not be. For small κ , its width is approximately κ times the off-data cost span, so whether it is large depends on the attainable κ , discussed in the next section.

4.3.1 The case of point-mass data distributions

We first calculate the full interval of expected costs at a fixed forward KL for data concentrated on a single successful outcome. Write $P = \delta_{x^*}$, where $c(x^*) = 0$. The KL constraint fixes $Q(x^*)$; we can choose how to distribute the remaining probability.

Proposition 4.3.1 (Cost interval for a point-mass reference). *Let $\mathcal{X}$ have at least two elements, fix $x^* \in \mathcal{X}$, and let $c : \mathcal{X} \rightarrow \mathbb{R}$ satisfy $c(x^*) = 0$. For $P = \delta_{x^*}$ and $\kappa \geq 0$,*

$$\{\mathbb{E}_Q[c] : D(P, Q) = \kappa\} = (1 - e^{-\kappa}) \left[\min_{x \neq x^*} c(x), \max_{x \neq x^*} c(x) \right]. \quad (4.21)$$

The endpoints are attained by placing all probability outside x^ on a minimizing or maximizing outcome, respectively.*

Proof. By algebra, the KL constraint is $-\log Q(x^*) = \kappa$. Thus every admissible law has the form

$$Q = e^{-\kappa} \delta_{x^*} + (1 - e^{-\kappa}) R, \quad R(x^*) = 0. \quad (4.22)$$

Since $c(x^*) = 0$, we have $\mathbb{E}_Q[c] = (1 - e^{-\kappa}) \mathbb{E}_R[c]$, where we are free to choose R subject to $R(x^*) = 0$. Point masses R at a cost minimizer or maximizer give the two endpoints. Mixtures of these point masses give every intermediate value. $\square$

In words, the result states that KL fixes the mass $1 - e^{-\kappa}$ assigned away from the demonstrated outcome; its average cost can be anywhere between the smallest and largest costs available there.

We note in passing that total variation records the same mass:

$$\|P - Q\|_{\text{TV}} = 1 - Q(x^*) = 1 - e^{-\kappa}.$$

Thus all models in the interval also have the same TV. The width of the cost interval tells us exactly how much task performance can vary while both distributional discrepancies remain unchanged.

4.3.2 Sequence examples: incorrect steps and final answers

We now ask how models with the same sequence KL can differ in two assessments of a generated solution: its number of incorrect steps and the correctness of its final answer. Applying [Proposition 4.3.1](#) to each cost gives the exact range of possible performance.

Now take $\mathcal{X} = \{0, 1\}^T$, with $T \geq 2$, and set $x^* = 0^T$. Think of this as a simplified model of generating a solution to a mathematical problem. For example, let a one record an error in a T -step argument, and let the final bit record whether the derived answer is incorrect. With this, there are two natural assessments of a derivation $x_{1:T} \in \{0, 1\}^T$:

$$c_{\text{cnt}}(x_{1:T}) = \sum_{t=1}^T x_t, \quad c_{\text{fin}}(x_{1:T}) = x_T. \quad (4.23)$$

The *error count* c_{cnt} measures how many steps are wrong. The *terminal cost* c_{fin} measures whether the final answer is wrong; earlier mistakes matter only through their effect on that answer. Under $P = \delta_{0^T}$, the data distribution has zero cost under either assessment.

Proposition 4.3.2 (Error count at fixed KL). *For $P = \delta_{0^T}$ and every $\kappa \geq 0$,*

$$\{\mathbb{E}_Q[c_{\text{cnt}}] : D(P, Q) = \kappa\} = [1 - e^{-\kappa}, T(1 - e^{-\kappa})], \quad (4.24)$$

where Q ranges over all laws on $\{0, 1\}^T$.

Proof. Outside 0^T , the smallest count is 1, attained at (say) 10^{T-1} , and the largest is T , attained at 1^T . Apply [Proposition 4.3.1](#). $\square$

At any positive KL, the largest possible expected count is T times the smallest. Even a sharp upper bound leaves this entire factor- T spread. A model that makes just one mistake on each erroneous sequence and a model that makes T mistakes can have exactly the same likelihood.

For terminal cost, an erroneous sequence can still end with a correct answer. This changes the lower endpoint to zero.

Proposition 4.3.3 (Terminal error at fixed KL). *For $P = \delta_{0^T}$, $T \geq 2$, and every $\kappa \geq 0$,*

$$\{\mathbb{E}_Q[c_{\text{fin}}] : D(P, Q) = \kappa\} = [0, 1 - e^{-\kappa}], \quad (4.25)$$

where Q ranges over all laws on $\{0, 1\}^T$.

Proof. Outside 0^T , terminal cost has minimum 0, attained at 10^{T-1} , and maximum 1, attained at 1^T . Apply [Proposition 4.3.1](#). $\square$

At the same KL, one generator can always give the correct final answer, while another has error probability $1 - e^{-\kappa}$. Here the cost range is fixed at $[0, 1]$, independently of the horizon.

The interval widths are governed by $1 - e^{-\kappa}$, which approaches one for large κ . For small κ ,

$$1 - e^{-\kappa} = \kappa + O(\kappa^2) \sim \kappa.$$

The expected count therefore ranges from approximately κ to $T\kappa$, and final-answer error ranges from zero to approximately κ . Small sequence KL gives small terminal error, but models with that same KL can differ throughout the permitted interval. At $\kappa = 1$, that interval extends from zero to approximately 0.632. Whether such a guarantee is useful depends on the task's required accuracy.

4.3.3 Attaining the cost bounds: recovery or persistence

We now construct two sequence laws that attain the lower and upper cost bounds in the preceding subsection. The construction in the proof of [Proposition 4.3.1](#) suggests taking

$$\begin{aligned} Q_- &= e^{-\kappa}\delta_{0T} + (1 - e^{-\kappa})\delta_{10^{T-1}}, \\ Q_+ &= e^{-\kappa}\delta_{0T} + (1 - e^{-\kappa})\delta_{1T}. \end{aligned} \tag{4.26}$$

The law Q_- attains the lowest cost, while Q_+ attains the highest cost for both cost functions considered in the preceding subsection, among all laws Q with $D(\delta_{0T}, Q) = \kappa$.

Generation under these laws has a similar form. Both start with $X_1 \sim \text{Bernoulli}(1 - e^{-\kappa})$. Under Q_- , one continues with $X_{t+1} = 0$; under Q_+ , one continues with $X_{t+1} = X_t$.

Under both laws, only the last bit is needed to generate the next one. This is the Markov property:

Definition 4.3.4 (Markov process). A finite-state discrete-time *stochastic process* is a collection of random variables $(X_t)_{t=1}^T$ taking values in a finite set $\mathcal{X}$. It has the *Markov property*, and is called a *Markov process*, if for every $t < T$ there is a map $K_t : \mathcal{X}^2 \rightarrow [0, 1]$ satisfying $\sum_{y \in \mathcal{X}} K_t(x, y) = 1$ for every $x \in \mathcal{X}$, such that, for every $y \in \mathcal{X}$, $\mathbb{P}$ -almost surely,

$$\mathbb{P}(X_{t+1} = y \mid X_1, \dots, X_t) = K_t(X_t, y).$$

The rule K_t is called the process's *transition kernel*. If the same kernel K is used at every time, the process is *time-homogeneous*.

After identifying $\mathcal{X}$ with $[d]$, we can represent K_t by a $d \times d$ nonnegative matrix whose (i, j) th entry is $K_t(i, j)$ and whose rows sum to one.

Both laws can be generated by time-homogeneous Markov processes. The following table gives one choice of transition kernel for each law, specified by $Q(X_{t+1} = 1 \mid X_t = x)$, along with the resulting costs:

	Q_-	Q_+
$Q(X_{t+1} = 1 \mid X_t = x)$	0	x
Expected error count	$1 - e^{-\kappa}$	$T(1 - e^{-\kappa})$
Final-answer error	0	$1 - e^{-\kappa}$

We can interpret the generation law of Q_- as recovering from an initial error, and that of Q_+ as persisting that error.

By the preceding results, the intervals also determine both moduli of continuity introduced in [Section 4.2.5](#). [Exercise 4](#) asks for these conversions and their interpretation: small expected error count forces small KL here, while perfect final answers permit arbitrarily large KL. In addition, [Exercise 5](#) compares the Pinsker and exact-TV bounds in a variant with an error opportunity at every step; the additional calculations make the same distinction between sharpening a guarantee and choosing a better generator.

4.4 What does likelihood fitting guarantee?

What does likelihood fitting guarantee about the generated solutions when data and computation are limited? We first ask how sequence length and parameter sharing affect learning accuracy, and why lower per-token loss need not mean better task performance. We then examine the accuracy required of the optimizer and how optimization error combines with finite-data error. Together, these comparisons identify what likelihood fitting controls and which differences between generators it leaves unresolved.

4.4.1 Repeated parameters and new parameters

Increasing the length of the training sequences gives us more observations, but fitting longer sequences requires predicting more tokens. Does the additional information compensate for the additional predictions? We compare cases where new positions reuse unknown parameters with cases where they introduce additional unknown parameters, then return to the resulting task-cost ranges.

Fix a data law P and a model class $\mathcal{Q}$ over infinite token sequences. Write $P_{1:T}, Q_{1:T}$ for their length- T prefix laws. At each length T , a learner receives n independent training sequences $X_{1:T}^{(1)}, \dots, X_{1:T}^{(n)}$ and fits a model $\hat{Q}^{(n,T)}$, a law over infinite token sequences. Define the *expected sequence KL for n training sequences of length T* as

$$\kappa_n(T) = \mathbb{E} \left[D(P_{1:T}, \hat{Q}_{1:T}^{(n,T)}) \right]. \quad (4.27)$$

The expectation averages over the training sample with law $P_{1:T}^{\otimes n}$ and any learner randomness. The fitted model can change with both n and T . We suppress the superscript (n, T) when the training sample size and length are clear, writing $\hat{Q} = \hat{Q}^{(n,T)}$. Subscripts such as $1:T$ indicate the prefix being evaluated. Unless specified otherwise, we keep n fixed as T increases.

For a fixed prediction task, additional training observations cannot worsen the best attainable statistical accuracy: a learner can ignore them. In particular, from n longer sequences it can keep only the shorter prefixes and run its original procedure. Here we also enlarge the prediction task. For an unchanged model Q , the KL chain rule in [Eq. \(4.5\)](#) gives

$$D(P_{1:T}, Q_{1:T}) \leq D(P_{1:T+1}, Q_{1:T+1}).$$

Learning from the additional observations can improve the fitted model. We ask whether that improvement compensates for the new prediction terms.

For models fitted by iterative optimization, exploiting better statistical accuracy also calls for a sufficiently small optimization error. Thus the appropriate fitting tolerance depends on both the amount of data and the sequence length. [Section 4.4.4](#) makes this balance explicit.

A simple estimation problem exhibits both possibilities. First suppose all tokens are independent Bernoulli(p) variables. This defines P . Estimate their common parameter using the smoothed frequency

$$\hat{p} = \frac{1 + \sum_{i=1}^n \sum_{t=1}^T X_t^{(i)}}{nT + 2}.$$

(Smoothing avoids the infinite expected KL of the ordinary empirical frequency; see [Exercise 7](#).) The fitted model $\hat{Q}^{(n,T)}$ is the independent Bernoulli bit model with parameter $\hat{p}$. For fixed $p \in (0, 1)$, the large-sample calculation gives

$$\mathbb{E}[(\hat{p} - p)^2] \sim \frac{p(1-p)}{nT}, \quad \kappa_n(T) = T \mathbb{E}[\text{kl}(p, \hat{p})] \longrightarrow \frac{1}{2n} \quad (T \rightarrow \infty). \quad (4.28)$$

Exercise 7 asks you to prove this and gives hints on the calculation. Note that each new position supplies another observation of the same unknown parameter. The gain in estimation accuracy compensates for the longer prediction task, giving $\lim_{T \rightarrow \infty} \kappa_n(T) = 1/(2n)$.

At the other extreme, let the tokens be independent with unrelated parameters p_t . In particular, for $t \geq 1$, X_t is Bernoulli with parameter p_t . Estimate each p_t separately from the n training sequences, using

$$\hat{p}_t = \frac{1 + \sum_{i=1}^n X_t^{(i)}}{n + 2}.$$

Each parameter still receives only n observations. For fixed n , each estimation risk remains positive; if $p_t \in [a, 1 - a]$ for a fixed $0 < a < 1/2$, their sum grows linearly with T . The large- n calculation identifies the leading coefficient: for fixed T and $p_1, \dots, p_T \in (0, 1)$,

$$\kappa_n(T) = \sum_{t=1}^T \mathbb{E}[\text{kl}(p_t, \hat{p}_t)] \sim \frac{T}{2n} \quad (n \rightarrow \infty). \quad (4.29)$$

Thus the leading estimation cost grows with the number of positions. The calculation describes the same fitting method in two model classes: one shared parameter versus one parameter per position.

Partial sharing. Fix a known partition $G_1, G_2, \dots$ of the positive integer positions, ordered by first occurrence. Tokens are independent, with $\mathbb{P}(X_t = 1) = p_j$ for $t \in G_j$. Let $d(T)$ be the number of groups represented among the first T positions, and let $m_j = |G_j \cap [T]|$. Pool all observations belonging to the same group and fit independent bits with the estimated probabilities

$$\hat{p}_j = \frac{1 + \sum_{i=1}^n \sum_{t \in G_j \cap [T]} X_t^{(i)}}{nm_j + 2}. \quad (4.30)$$

The resulting sequence KL grows with the number of unknown parameters: each contributes approximately $1/(2n)$.

Proposition 4.4.1 (Sequence KL under parameter sharing). *In the grouped independent-bit model, fit each probability by the pooled smoothed frequency from n independent length- T training sequences. Fix $0 < a < 1/2$ and suppose $p_j \in [a, 1 - a]$. Then*

$$\kappa_n(T) \asymp_a \frac{d(T)}{n}, \quad n, T \geq 1. \quad (4.31)$$

Here $\asymp_a$ means upper and lower bounds by positive constant multiples, with constants depending only on a . If $n \min_{1 \leq j \leq d(T)} m_j \rightarrow \infty$, then more precisely

$$\kappa_n(T) \sim \frac{d(T)}{2n}. \quad (4.32)$$

Each independently estimated parameter therefore contributes approximately $1/(2n)$, regardless of how many positions use it. To see the cancellation, group j supplies nm_j training observations and contributes m_j terms to the test-sequence KL:

$$\kappa_n(T) = \sum_{j=1}^{d(T)} m_j \mathbb{E}[\text{kl}(p_j, \hat{p}_j)] \approx \sum_{j=1}^{d(T)} m_j \frac{1}{2nm_j} = \frac{d(T)}{2n}.$$

The condition in the proposition says that every pooled sample size becomes large, so this approximation is accurate across all groups. [Exercise 7](#) develops the proof; [Appendix 4.D.2](#) gives the proof.

The two extremes are $d(T) = 1$ and $d(T) = T$. Intermediate sharing patterns give intermediate growth of the estimation cost. For example, using one new parameter for each successive block of lengths $1, 2, 3, \dots$ gives $d(T)$ of order $\sqrt{T}$. A fixed finite collection of known candidate models offers another possibility: exact identification can become sufficiently reliable that $\kappa_n(T)$ tends to zero (even with n fixed). [Exercise 10](#) asks for this calculation with two Bernoulli candidates.

These comparisons keep the number of independent sequences n fixed, allowing the training-token budget nT to grow. If instead we fix $L = nT$, write $\kappa'_L(T) := \kappa_{L/T}(T)$. In the fully shared model, substituting $n = L/T$ and $d(T) = 1$ into [Eq. \(4.32\)](#) gives

$$\kappa'_L(T) \sim \frac{1}{2(L/T)} = \frac{T}{2L} \quad (L \rightarrow \infty). \quad (4.33)$$

Thus, at a fixed large training-token budget, increasing T worsens expected sequence KL, which grows approximately linearly. [Exercise 8](#) also asks for the unshared comparison.

For the class with $p_j \in [a, 1 - a]$, the order $d(T)/n$ is also *minimax optimal*: no learner improves its worst-case dependence on n and $d(T)$; see the [bibliographical notes](#).

Proposition 4.4.2 (Learning rates and task-cost intervals). *For any d with $1 \leq d(T) \leq T$, there exist a data law, terminal and counting costs, and estimator families such that, for all $n \geq 1, T \geq 2$, the estimators have common expected sequence KL $\kappa_n(T) \asymp d(T)/n$ and their expected costs range over exactly*

$$\left[0, 1 - e^{-\kappa_n(T)/2}\right] \quad \text{and} \quad \left[1 - e^{-\kappa_n(T)/2}, T(1 - e^{-\kappa_n(T)/2})\right],$$

respectively.

To get this result, combine the grouped Bernoulli construction with [Section 4.3](#), using tokens $X_t = (U_t, V_t)$, with costs computed from $V_{1:T}$. [Exercise 9](#) develops the proof.

What should we expect for transformers? In a mathematical solution, many steps apply familiar rules, while later steps may require combining definitions and intermediate results across a long prefix. The statistical question is how much useful sharing training discovers. The computational question is whether the architecture can carry out the required combinations with its available depth, width, and memory. The Bernoulli examples isolate the statistical benefit when the sharing pattern is known. [Exercise 11](#) studies a separate obstruction: a short copying rule whose required memory grows.

For mathematical reasoning, a plausible working hypothesis is substantial reuse together with additional demands as problems grow. The grouped model gives one way to quantify that hypothesis; the number $d(T)$ counts its independently estimated probabilities. Which sharing is useful and executable in a transformer is a question about the task and the learned computation. [Exercise 12](#) studies a missed parity dependency as an exercise in model-class restrictions and optimization accuracy. Lecture 18 develops the statistical horizon comparison further.

4.4.2 The pitfalls of reporting per-token log loss

As sequence length grows, log loss per token can vanish while final-answer error stays constant. For the all-zero law P and persistent generator Q_+ from [Eq. \(4.26\)](#), with fixed $\kappa > 0$,

$$\frac{D(P_{1:T}, (Q_+)_{1:T})}{T} = \frac{\kappa}{T} \rightarrow 0, \quad \mathbb{E}_{(Q_+)_{1:T}}[c_{\text{fin}}] = 1 - e^{-\kappa}. \quad (4.34)$$

The lower per-token loss comes entirely from dividing the same sequence loss by a larger T . When comparing different sequence lengths, lower per-token log loss alone is therefore insufficient evidence of better final-answer accuracy.

4.4.3 Likelihood fitting at finite accuracy

We use loss per predicted token as the optimization objective. For n observed sequences, define empirical *per-token* log loss

$$\hat{L}(Q) = \frac{1}{nT} \sum_{i=1}^n \sum_{t=1}^T -\log Q(X_t^{(i)} | X_{<t}^{(i)}).$$

This averages the losses of the nT token predictions. We use this normalization despite the warning in [Section 4.4.2](#): per-token loss can improve as sequence length grows without task performance improving. There are two reasons for using it. First, when token losses have comparable scales, normalization removes the explicit factor of T that would otherwise multiply the objective and its gradient. For gradient descent, this avoids a corresponding adjustment of the stepsize and helps us use common optimization settings across sequence lengths. Second, log loss is conventionally reported per token.³

When an optimizer is run with finite compute, it generally does not return a model with the lowest possible empirical loss. The *optimization error* of its returned model $\hat{Q}$ is

$$E(\hat{Q}) := \hat{L}(\hat{Q}) - \inf_{Q \in \mathcal{Q}} \hat{L}(Q). \quad (4.35)$$

We say that $\hat{Q}$ satisfies the *optimization tolerance* $\eta \geq 0$ if $E(\hat{Q}) \leq \eta$, equivalently,

$$\hat{L}(\hat{Q}) \leq \inf_{Q \in \mathcal{Q}} \hat{L}(Q) + \eta. \quad (4.36)$$

Establishing whether a returned model satisfies this tolerance is a *certification question*. For example, suppose $\mathcal{Q} = \{Q_\theta : \theta \in \mathbb{R}^d\}$ and $f(\theta) = \hat{L}(Q_\theta)$ is differentiable and strongly convex with known parameter $\mu > 0$. Then

$$E(Q_\theta) \leq \frac{\|\nabla f(\theta)\|_2^2}{2\mu},$$

so $\|\nabla f(\theta)\|_2 \leq \sqrt{2\mu\eta}$ certifies the tolerance ([Exercise 13](#)). A small gradient alone gives no such guarantee for a general nonconvex loss. We can still study the consequences of positive optimization error even when its size is unknown to the optimizer.

Because of the normalization by T , a per-token tolerance η only guarantees a per-sequence tolerance $T\eta$. Our earlier cost bounds motivate controlling the per-sequence error. Maintaining a fixed per-sequence tolerance as T increases requires increasingly accurate per-token optimization, which may require more compute. Whether a fixed number of passes through the dataset suffices depends on the optimization problem and how the number of updates scales with T .⁴

Positive optimization error can translate into poor task performance, just as sequence-level KL error can. Since η is measured per token, the relevant sequence-level quantity is $T\eta$. The simplest illustration uses the deterministic data law $P = \delta_{0T}$ from [Section 4.3.3](#): empirical and population log loss coincide, so there is no statistical error to account for. Reusing that example gives the following result.

Proposition 4.4.3 (Different task costs at the same optimization tolerance). *Let $T \geq 2$. Take the laws Q_-, Q_+ in Eq. (4.26) with $\kappa = T\eta$, and let the model class contain P, Q_-, Q_+ . For every sample of n sequences from $P = \delta_{0^T}$,*

$$\begin{aligned}\widehat{L}(P) &= 0, & \widehat{L}(Q_-) &= \widehat{L}(Q_+) = \eta, \\ \mathbb{E}_{Q_-}[c_{\text{fin}}] &= 0, & \mathbb{E}_{Q_+}[c_{\text{fin}}] &= 1 - e^{-T\eta}.\end{aligned}\tag{4.37}$$

Both generators satisfy Eq. (4.36).

Proof. Each data sequence is 0^T . Both generators assign it probability $e^{-T\eta}$, so their loss per token is η . Their terminal costs follow from Eq. (4.26). $\square$

For small $T\eta$, the permitted final-answer error is approximately $T\eta$. For example, $\eta = 10^{-3}$ and $T = 1000$ permit anything from zero terminal error to about 0.632. More demonstrations of 0^T leave this choice unchanged.

If the class consists of Q_-, Q_+ at a fixed $\kappa > 0$, then the data law P is outside the class. Both generators minimize empirical and population log loss, yet their terminal costs are zero and $1 - e^{-\kappa}$. Thus, in addition to optimization error, *approximation error* can also leave a consequential choice between equally fitted models. Exercise 14 develops this comparison with repeated errors, and Exercise 12 studies a missed dependency in a parity task.

4.4.4 What does finite-sample likelihood fitting guarantee?*

Likelihood fitting gives a positive statistical guarantee on task performance. The following stylized result bounds the task cost of choosing a model with nearly minimal empirical log loss from a finite class. Allowing optimization slack, as in the previous subsection, lets us ask how optimization effort should depend on sample size.

Theorem 4.4.4 (Approximate maximum likelihood controls cost). *Let $\mathcal{Q}$ be a finite class containing P , and observe n independent sequences with law P . Let $0 \leq c \leq C_{\max}$. Suppose $\widehat{Q} \in \mathcal{Q}$ satisfies Eq. (4.36) for $\eta \geq 0$. For $0 < \delta < 1$, put*

$$b = \frac{2 \log(|\mathcal{Q}|/\delta)}{n} + T\eta.$$

With probability at least $1 - \delta$,

$$d_c(P, \widehat{Q}) \leq 2\sigma_P(c)\sqrt{b} + C_{\max}b.\tag{4.38}$$

In particular, if $\mathbb{E}_P[c] = 0$, then $\mathbb{E}_{\widehat{Q}}[c] \leq C_{\max}b$.

The proof is in Appendix 4.D.1. For zero-cost data and $0 \leq c \leq 1$, the theorem separates the two contributions to its guarantee:

$$\mathbb{E}_{\widehat{Q}}[c] \leq \underbrace{\frac{2 \log(|\mathcal{Q}|/\delta)}{n}}_{\text{statistical contribution}} + \underbrace{T\eta}_{\text{optimization contribution}}.\tag{4.39}$$

At fixed model class and confidence level, the statistical contribution is of order $1/n$ and remains in the bound even with exact optimization. If optimization effort is insufficient, the optimization contribution $T\eta$ may dominate. To keep it no larger than the statistical contribution, it suffices to achieve

$$\eta \leq \frac{2 \log(|\mathcal{Q}|/\delta)}{nT}.\tag{4.40}$$

Substitution into Eq. (4.39) gives $\mathbb{E}_{\hat{Q}}[c] \leq 4 \log(|\mathcal{Q}|/\delta)/n$, with probability at least $1 - \delta$. *Optimizing to statistical accuracy* means keeping the optimization contribution on the scale of that statistical term. Even at fixed sequence length, retaining this statistical rate calls for a decreasing per-token tolerance as the number of examples grows.

The example in Proposition 4.4.3 shows that the optimization contribution has the correct order when $T\eta$ is small.

Lecture 18 develops this comparison for imitation learning. The comparison in Section 4.4.1 fixes infinite-sequence models and varies the length of the observed prefixes. Its Bernoulli examples isolate parameter sharing; the present finite-class theorem also accounts for optimization tolerance. The number of sequences and the token budget specify different learning comparisons.

4.5 How can we improve task performance?

Likelihood fitting can leave models with similar losses but very different task performance. How can training favor the more successful models? We examine two approaches: changing the objective to assess the consequences of generation, and giving the model structure that supports recovery from errors. Recovery then raises a further question: which information should be forgotten, and which must be preserved?

4.5.1 Evaluate consequences of generation

Direct cost feedback. An evaluator for the task can compare the models through what they generate. On the terminal task, it assigns Q_- expected cost zero and Q_+ expected cost $1 - e^{-\kappa}$. This directly supplies the distinction that their identical likelihoods leave unresolved. If cost can be evaluated on new generations, the objective is

$$\min_{\theta} \mathbb{E}_{Q_{\theta}}[c]. \quad (4.41)$$

The additional information is an assessment of generated sequences, including ones absent from the demonstrations. Lecture 19 develops reinforcement learning with this feedback, including the policy-gradient method for optimizing such an objective. Lectures 20–21 study preference feedback and optimization with learned evaluators.

Predict further ahead. The existing demonstrations can also distinguish Q_- and Q_+ through a different prediction question. Both have sequence log loss κ on 0^T , but predicting its final token from the empty prefix gives

$$-\log Q_-(X_T = 0) = 0, \quad -\log Q_+(X_T = 0) = \kappa. \quad (4.42)$$

This loss prefers the model that recovers. The target is observed in the demonstration; predicting it without the intervening tokens evaluates the model’s longer consequences.

A further-ahead prediction loss asks the model to predict X_{t+k} from $X_{\leq t}$, before the intervening tokens are supplied. One possible objective adds

$$\mathcal{L}_k(\theta) = - \sum_{t=0}^{T-k} \log Q_{\theta}(X_{t+k} \mid X_{\leq t}) \quad (4.43)$$

to next-token loss. This evaluates a longer consequence of the model’s conditional probabilities.

For a general autoregressive model, however,

$$Q_\theta(X_{t+k} = a \mid x_{\leq t}) = \sum_{y_{1:k-1} \in [N]^{k-1}} \left[\prod_{j=1}^{k-1} Q_\theta(y_j \mid x_{\leq t}, y_{<j}) \right] \cdot Q_\theta(a \mid x_{\leq t}, y_{1:k-1}). \quad (4.44)$$

The intervening tokens are unobserved in this prediction problem. Naive evaluation enumerates N^{k-1} possibilities. Using the intervening *observed* tokens would return to a one-step conditional. A separate prediction head can be fitted directly to the observed input–target pairs, but requiring it to match the marginal in Eq. (4.44) brings back this computation. [Exercise 15](#) compares naive enumeration with the savings available from Markov structure.

Fit reverse KL. Reverse KL uses concentration of the cost under P to control generated cost ([Theorem 4.2.8](#)). To assess it as a fitting objective, we ask which generators it accepts and what information is needed to evaluate it.

Forward KL averages conditional discrepancies at data prefixes. Reversing its arguments averages them at generated prefixes:

$$D(Q, P) = \sum_{t=1}^T \mathbb{E} \left[D(Q(\cdot \mid \tilde{X}_{<t}), P(\cdot \mid \tilde{X}_{<t})) \right], \quad \tilde{X}_{1:T} \sim Q. \quad (4.45)$$

The generator’s own erroneous histories now contribute.

Successful generation and the data’s support. For $P = \delta_{0T}$, every law with finite reverse KL equals P . In particular, at $\kappa > 0$, the two laws satisfy

$$\mathbb{E}_{Q_-}[c_{\text{fin}}] = 0, \quad D(Q_-, P) = D(Q_+, P) = +\infty.$$

Reverse KL excludes both the persistent error and the successful recovery path 10^{T-1} . It requires staying within the support of the demonstrated distribution; direct task feedback accepts any successful sequence.

This support requirement can still allow many successful generators. Let P be uniform on a set $\mathcal{A}$ of M zero-cost sequences. Every Q supported on $\mathcal{A}$ satisfies

$$D(Q, P) = \log M - H(Q) \leq \log M, \quad \mathbb{E}_Q[c] = 0.$$

A generator that always selects one demonstrated solution has zero task cost, even though it does not reproduce the data frequencies.

What the two directions distinguish. Reverse KL penalizes implausible generated behavior; forward KL penalizes missing data-supported behavior. Two constructions make the distinction precise.

Proposition 4.5.1 (Adding bad outcomes and omitting good ones). *Let P, R be probability laws on the finite domain $\mathcal{X}$.*

(i) *For $0 < \varepsilon < 1$, the mixture $Q = (1 - \varepsilon)P + \varepsilon R$ satisfies*

$$D(P, Q) \leq -\log(1 - \varepsilon), \quad \mathbb{E}_Q[c] = (1 - \varepsilon)\mathbb{E}_P[c] + \varepsilon\mathbb{E}_R[c]. \quad (4.46)$$

(ii) For an event $B \subset \mathcal{X}$ with $0 < P(B) < 1$, define the conditional law $Q = P(\cdot | B)$ on the same domain by

$$Q(x) := \frac{P(x)\mathbb{I}\{x \in B\}}{P(B)}, \quad x \in \mathcal{X}.$$

Then

$$D(Q, P) = -\log P(B), \quad D(P, Q) = +\infty. \quad (4.47)$$

Proof. For (i), $Q(x) \geq (1 - \varepsilon)P(x)$ bounds each log ratio under P ; expectation is linear in the law. For (ii), write

$$P = P(B)Q + (1 - P(B))P(\cdot | B^c).$$

The components have disjoint supports, so the sharpness calculation following [Proposition 4.2.4](#) gives $D(Q, P) = -\log P(B)$. Also, $Q(B^c) = 0 < P(B^c)$, so $D(P, Q) = +\infty$. $\square$

In (i), the forward-KL bound is independent of how costly the added outcomes are. If the data have zero cost and R always incurs cost $C_{\max}$, the generated cost is $\varepsilon C_{\max}$. In (ii), discarding half the data probability costs only $\log 2$ in reverse KL. Thus a moderate reverse KL can coexist with substantial loss of coverage. For the present zero-cost task, discarding successful outputs can preserve perfect task performance. A requirement to reproduce those outputs would need to be part of the task objective.

The information needed for reverse-KL fitting. What information would let us minimize this discrepancy?

Controlling reverse KL poses an additional learning problem:

$$D(Q_\theta, P) = \mathbb{E}_{Q_\theta}[\log Q_\theta - \log P]. \quad (4.48)$$

Generating a sequence lets us evaluate its model probability. Its probability under the unknown data law is another matter. Using only the empirical data law would give infinite reverse KL to every generator that assigns positive mass outside the training set. Target-density or density-ratio estimation requires further assumptions about how the data generalize.

There is also a statistical difficulty hidden in the support condition. Without prompts, replaying complete observed sequences gives a law supported on P . With externally supplied prompts $U \sim \mu$, the model must answer new prompts as well. If the correct answer is a deterministic bit $f(U)$, finite reverse KL requires answering correctly at *every* prompt of positive μ -probability.

Rare prompts can make support recovery expensive.* Consider two known candidate answer rules that disagree on prompts of total probability α . From n independent demonstrations, the smallest worst-case probability of returning a model with infinite reverse KL is exactly

$$\frac{1}{2}(1 - \alpha)^n.$$

Until a distinguishing prompt occurs, the data give no information about which rule is correct. In contrast, a predictor that randomizes equally between the two answers on those prompts has forward KL $\alpha \log 2$ and error probability $\alpha/2$. [Appendix 4.D.3](#) states and proves this result, including an example with $\alpha = 2^{-m}$ on prompts containing $2m$ bits. Thus even a two-model class can require exponentially many random demonstrations to meet the reverse-KL support requirement.

[Appendix 4.E](#) explains how variational methods exploit a target density known up to normalization, and how a known cost can be used to define such a target.

4.5.2 Recovery at unchanged likelihood

Recovery limits how errors affect later generation. If the model's transitions contract discrepancies between state distributions by a factor $\rho < 1$, the discrepancy at the next step is at most ρ times the current discrepancy plus the new prediction error. Repeated errors therefore contribute a geometric sum, giving a bound at every time. We first establish this bound, then illustrate how recovery can improve task performance at unchanged likelihood.

A general error-propagation bound. For a law μ and kernel K , write

$$(\mu K)(y) = \sum_x \mu(x) K(x, y).$$

Let P, Q be Markov laws with kernels $K_t, \hat{K}_t$, respectively, for the transition from time t to time $t + 1$. Their marginal laws $\mu_t = P(X_t \in \cdot)$ and $\nu_t = Q(X_t \in \cdot)$ obey $\mu_{t+1} = \mu_t K_t$ and $\nu_{t+1} = \nu_t \hat{K}_t$.

Theorem 4.5.2 (Local error and forgetting). *Suppose that, for $1 \leq t < T$, the fitted kernels satisfy*

$$\|\alpha \hat{K}_t - \beta \hat{K}_t\|_{\text{TV}} \leq \rho \|\alpha - \beta\|_{\text{TV}} \quad \text{for all laws } \alpha, \beta, \quad 0 \leq \rho \leq 1, \quad (4.49)$$

and their local errors under the true process satisfy

$$\sum_x \mu_t(x) \|K_t(x, \cdot) - \hat{K}_t(x, \cdot)\|_{\text{TV}} \leq \varepsilon_{t+1}. \quad (4.50)$$

Then $e_t = \|\mu_t - \nu_t\|_{\text{TV}}$ satisfies

$$\begin{aligned} e_{t+1} &\leq \rho e_t + \varepsilon_{t+1}, \\ e_T &\leq \rho^{T-1} e_1 + \sum_{t=2}^T \rho^{T-t} \varepsilon_t. \end{aligned} \quad (4.51)$$

If $\rho < 1$, $e_1 \leq \varepsilon$, and $\varepsilon_t \leq \varepsilon$ for all $2 \leq t \leq T$, then $e_t \leq \varepsilon/(1 - \rho)$ at every time. Consequently, every $f : \mathcal{X} \rightarrow [0, 1]$ satisfies

$$\max_{1 \leq t \leq T} |\mathbb{E}_P[f(X_t)] - \mathbb{E}_Q[f(X_t)]| \leq \frac{\varepsilon}{1 - \rho}. \quad (4.52)$$

Proof. Insert $\mu_t \hat{K}_t$ between the two next-state laws:

$$\|\mu_t K_t - \nu_t \hat{K}_t\|_{\text{TV}} \leq \|\mu_t K_t - \mu_t \hat{K}_t\|_{\text{TV}} + \|\mu_t \hat{K}_t - \nu_t \hat{K}_t\|_{\text{TV}} \leq \varepsilon_{t+1} + \rho e_t.$$

Iteration gives Eq. (4.51). Under the uniform error bounds, $e_t \leq \varepsilon \sum_{j=0}^{t-1} \rho^j \leq \varepsilon/(1 - \rho)$. Apply Proposition 4.2.2 to f at each time and take the maximum. $\square$

The factor ρ^{T-t} measures how much the subsequent transitions attenuate an error introduced at time t . An initial error can decay for $T - 1$ steps; a final prediction error contributes without attenuation. Repeated local errors contribute a geometric sum.

For a finite-state kernel, the smallest possible contraction factor is

$$\rho(\hat{K}) = \max_{x, x'} \|\hat{K}(x, \cdot) - \hat{K}(x', \cdot)\|_{\text{TV}}. \quad (4.53)$$

Exercise 16 proves this characterization.

Recovery in the binary example. We now improve the persistent law Q_+ by giving each error a chance to be corrected. This changes the generated cost while leaving its likelihood on the data unchanged.

Fix $P = \delta_{0^T}$ and $\kappa \geq 0$. Let Q_r use the same first-bit distribution as Q_- and Q_+ , followed by

$$Q_r(X_1 = 1) = 1 - e^{-\kappa},$$

$$Q_r(X_{t+1} = 1 \mid X_t = 0) = 0, \quad Q_r(X_{t+1} = 0 \mid X_t = 1) = r, \quad 1 \leq t < T,$$

where $0 \leq r \leq 1$. Once an error is corrected, all later bits are zero. The choices $r = 0$ and $r = 1$ give Q_+ and Q_- , respectively.

Proposition 4.5.3 (Recovery reduces cost at unchanged likelihood). *For this family and $T \geq 2$,*

$$D(P, Q_r) = \kappa, \quad \mathbb{E}_{Q_r}[c_{\text{fin}}] = (1 - e^{-\kappa})(1 - r)^{T-1}. \quad (4.54)$$

For $r > 0$,

$$\mathbb{E}_{Q_r}[c_{\text{cnt}}] = (1 - e^{-\kappa}) \frac{1 - (1 - r)^T}{r} \leq \frac{1 - e^{-\kappa}}{r}. \quad (4.55)$$

At $r = 0$, the expected count is $T(1 - e^{-\kappa})$.

Proof. The all-zero path has probability $e^{-\kappa}$, independently of r . Also,

$$Q_r(X_t = 1) = (1 - e^{-\kappa})(1 - r)^{t-1},$$

since the first bit must be wrong and each subsequent correction must fail. Taking $t = T$ gives terminal error; summing over t gives the expected count. $\square$

At a fixed positive recovery probability, terminal error decays geometrically and expected error count stays bounded as the horizon grows. An erroneous run has geometric duration with mean $1/r$ before truncation at the horizon. Likelihood evaluates the probability of the initial error; the recovery rule determines how costly that error becomes.

For Q_r , the two rows differ by $1 - r$, so its kernels after the first token have contraction factor $1 - r$. The initial discrepancy is $e_1 = 1 - e^{-\kappa}$, and all subsequent local errors under P are zero. The theorem therefore gives

$$e_T \leq (1 - r)^{T-1}(1 - e^{-\kappa}),$$

which is exactly the terminal cost in Eq. (4.54). Exercise 16 adds fresh errors at every step and computes the resulting long-run error level. The next question is how to obtain recovery while retaining information the task needs.

4.5.3 What should be forgotten?

Random resets give a simple mathematical way to enforce forgetting. For any kernel K and reset law π_{reset} , the kernel

$$K_r(x, \cdot) = (1 - r)K(x, \cdot) + r\pi_{\text{reset}}(\cdot)$$

has contraction factor at most $1 - r$. A reset makes the two initial states use the same distribution. But forgetting everything can erase exactly the information a task needs. A proof generator must retain the theorem being proved and useful facts established earlier.

Limited context and contraction are different properties. The rule $X_{t+1} = X_t$ uses only one token of context and preserves its initial bit forever. During generation, recent tokens can keep transmitting old information. Restricting a transformer's accessible prefix therefore does not establish Eq. (4.49).

For prompting, a sliding window creates a more immediate difficulty: a fact in the original prompt can leave the accessible context before the model has carried it forward. Subsequent predictions then have no direct access to it. Generated reminders can preserve information, as in the copying example, but can also introduce errors into what is retained. Keeping the original prompt or verified task information separately provides a more reliable source than repeatedly reconstructing it from generated text.

A more useful design separates persistent task information M from a revisable working state R_t :

$$S_t = (M, R_t), \quad R_{t+1} = g_M(R_t, \Xi_{t+1}). \quad (4.56)$$

Conditional on the same M , the working dynamics can forget perturbations while M retains the problem and trusted information. For example, in proof generation this suggests retaining the problem statement and checked lemmas while restarting a failed attempt. The mathematical question is which state differences can be made contractive without destroying information the task needs. An explicit state-space model gives a way to express this separation.

4.6 Representing the information that should persist

A long computation may need an early definition, a constraint, or a fact established many steps ago. A fixed number of recent observations can be an inconvenient representation of this information. A *state-space model* instead gives an explicit state and a rule for maintaining it.

One nonlinear stochastic model is

$$S_{t+1} = f_\theta(S_t, \Xi_{t+1}), \quad X_{t+1} = g_\theta(S_{t+1}, \Upsilon_{t+1}). \quad (4.57)$$

The initial law of S_0 and the laws of the noise arguments complete the model. Assume here that the noise pairs are independent across time, that the two noises within a pair are independent, and that they are independent of S_0 . The state process is Markov. The observation process is a *hidden Markov process*: its dependence is mediated by an unobserved, or *latent*, state. Its observations need not themselves be Markov.

A state can preserve a component for as long as needed by copying it. For instance, one coordinate can retain a bit learned at the beginning of the sequence, while others store later facts or working information. This permits long memory with a fixed update rule. Storing k independent bits requires at least 2^k distinct states; a vector of k binary coordinates is a compact way to describe that large state space. Selecting what to store is part of the modeling problem.

These are old questions in sequence modeling. In engineering, state coordinates may describe position, velocity, temperature, or other physical quantities, and physical knowledge suggests their evolution and measurement equations. In language tasks, a useful state representation is less directly specified. In both cases, the state assumption relates many observations through the same evolution rule.

4.6.1 The statistical cost of hidden states

The learner sees observations, so maximum likelihood must sum over the hidden explanations of those observations. For a finite HMM, write $J_t \in [d]$ for its hidden-state label and set

$$A_{ij} = \mathbb{P}(J_{t+1} = i \mid J_t = j), \quad C_{aj} = \mathbb{P}(X_t = a \mid J_t = j).$$

Columns of $A \in \mathbb{R}^{d \times d}$ are next-state distributions; columns of $C \in \mathbb{R}^{N \times d}$ are emission distributions. Both matrices are column-stochastic.⁵ With initial distribution π , the likelihood is

$$Q_\theta(X_{1:T}) = \sum_{j_{0:T} \in [d]^{T+1}} \pi_{j_0} \prod_{t=1}^T A_{j_t j_{t-1}} C_{X_t j_t}, \quad \theta = (\pi, A, C). \quad (4.58)$$

Naive evaluation visits exponentially many trajectories. HMM structure allows the sum to be computed in $O(Td^2)$ operations by a forward recursion; [Exercise 18](#) asks for this algorithm. When $d = 2^k$, even this dependence on the state count can be expensive.

If hidden states were observed, transition and emission counts would give direct frequency estimates. With hidden states, the corresponding posterior expected counts depend on the current parameters. The sum inside $-\log Q_\theta(X_{1:T})$ couples the transition and emission parameters. For continuous states, the sum becomes an integral over trajectories. Variational approximation to this likelihood is outlined in [Appendix 4.E.3](#).

There is also an important computationally simpler use of a recurrent state. Suppose a predictor fixes R_0 and computes

$$R_t = h_\theta(R_{t-1}, X_t), \quad Q_\theta(\cdot \mid X_{\leq t}) = q_\theta(R_t).$$

The observed tokens and parameters determine every R_t . Teacher-forced likelihood is evaluated and differentiated through that recursion. The issue requiring the sum in [Eq. \(4.58\)](#) is stochastic unobserved state trajectories; a deterministically computed memory is available during likelihood evaluation.

4.6.2 Discrete processes in vector spaces

To connect this discussion to linear models, represent each token by its one-hot vector:

$$Z_t = e_{X_t} \in \mathbb{R}^N, \quad \mathcal{F}_t = \sigma(X_1, \dots, X_t).$$

The notation $\mathcal{F}_t$ records the information in the observations through time t .

Proposition 4.6.1 (Conditional probabilities as a vector mean). *For every discrete process, define*

$$P_t = \mathbb{E}[Z_{t+1} \mid \mathcal{F}_t], \quad E_{t+1} = Z_{t+1} - P_t.$$

Then

$$Z_{t+1} = P_t + E_{t+1}, \quad \mathbb{E}[E_{t+1} \mid \mathcal{F}_t] = 0, \quad (P_t)_a = \mathbb{P}(X_{t+1} = a \mid \mathcal{F}_t). \quad (4.59)$$

Proof. Coordinate a of Z_{t+1} is $\mathbb{I}\{X_{t+1} = a\}$. Its conditional expectation is the corresponding conditional probability. Subtracting that expectation gives the residual identity. $\square$

A noise sequence with this conditional-mean-zero property is called a *martingale difference sequence*. The conditional mean P_t is itself random because it depends on the observed history. It contains the entire next-token distribution. Writing $P_t = \phi_t(X_{1:t})$ gives a general autoregression

$$Z_{t+1} = \phi_t(X_{1:t}) + E_{t+1}.$$

Every integrable vector process has this conditional-mean-plus-noise representation. Squared prediction loss also has a familiar meaning here: for a probability-vector report q , $\|e_{X_{t+1}} - q\|_2^2$ is the Brier loss of [Lecture 3](#).

For finite Markov states, the conditional mean becomes linear after one-hot encoding. In the HMM above, set $S_t = e_{J_t} \in \mathbb{R}^d$ and $Z_t = e_{X_t} \in \mathbb{R}^N$. Then

$$S_{t+1} = AS_t + W_{t+1}, \quad Z_{t+1} = CS_{t+1} + V_{t+1}, \quad (4.60)$$

where the residuals satisfy

$$\mathbb{E}[W_{t+1} \mid \mathcal{G}_t] = 0, \quad \mathbb{E}[V_{t+1} \mid \mathcal{G}_t, S_{t+1}] = 0, \quad \mathcal{G}_t = \sigma(S_0, \dots, S_t, Z_1, \dots, Z_t). \quad (4.61)$$

Indeed, columns of A and C are precisely the relevant conditional means. Subtract them to obtain W_{t+1} and V_{t+1} . These residual distributions depend on the hidden state. Their conditional-mean properties, rather than independence, make the linear representation possible.

Thus a discrete HMM has linear state and observation equations with martingale noise. A large finite state space becomes a high-dimensional one-hot vector. For an arbitrary process, the entire history can serve as a state, but the number of such states grows with time. An economical fixed-dimensional linear model is a substantive structural assumption.

4.7 Learning dynamics through observable prediction

System identification asks for state-transition and observation equations from observed sequences. We have just seen why a direct latent-state likelihood can be difficult to fit. Linear models provide another route: first learn to predict from observations, then optionally recover a compact state representation.

In Eq. (4.60), now allow $S_t \in \mathbb{R}^d$ and $Z_t \in \mathbb{R}^m$, with $A \in \mathbb{R}^{d \times d}$ and $C \in \mathbb{R}^{m \times d}$. The matrix A maps state coordinates to next-state coordinates; each row of C specifies a measured coordinate. This is a *linear dynamical system* (LDS). It includes engineering models with real measurements as well as the one-hot construction above, with different noise laws.

For example, consider motion sampled at intervals Δ . Use position and velocity as the two state coordinates. The constant-velocity model has matrices

$$A = \begin{pmatrix} 1 & \Delta \\ 0 & 1 \end{pmatrix}, \quad C = (1, 0).$$

The observation measures position; the hidden velocity influences all future positions. Process and measurement noises account for departures from this idealized motion and errors in the observations.

In what follows, vector norms are Euclidean, the matrix norm $\|A\|$ is the induced operator norm, and $\|A\|_F$ is the Frobenius norm. The spectral radius $\rho(A)$ is the largest modulus of an eigenvalue. We call a matrix *stable* when $\rho(A) < 1$.

If remote observations have little predictive effect, use a *linear autoregression of order p* :

$$\hat{Z}_{t+1} = \sum_{j=1}^p B_j Z_{t+1-j} = B \Psi_t, \quad \Psi_t = \begin{pmatrix} Z_t \\ Z_{t-1} \\ \vdots \\ Z_{t-p+1} \end{pmatrix}, \quad B = (B_1, \dots, B_p). \quad (4.62)$$

Here $\Psi_t \in \mathbb{R}^{mp}$, each $B_j \in \mathbb{R}^{m \times m}$, and $B \in \mathbb{R}^{m \times mp}$. The word autoregression now makes a *linear conditional-prediction assumption*, beyond the probability factorization in Eq. (4.2).

Proposition 4.7.1 (Observable fitting is convex). *Given the observed sequence, the fitting problem*

$$\hat{B} \in \arg \min_B \sum_{t=p}^{T-1} \|Z_{t+1} - B\Psi_t\|_2^2 \quad (4.63)$$

is convex. All its features and targets are observed.

Proof. For each Ψ_t , the prediction $B\Psi_t$ is linear in B , and squared Euclidean distance is convex. $\square$

This is the same data-access advantage as next-token prediction: the observations supply both the context and the answer. For comparison, even a deterministic hidden trajectory with Gaussian measurement noise leads to the objective

$$\min_{A,C,s} \sum_{t=1}^T \|Z_t - CA^{t-1}s\|_2^2,$$

where unknown quantities multiply and A appears through its powers.

What might we lose by fitting the lag coefficients separately? A finite-dimensional state model shares its dynamics across all lags. Below, its history coefficients take the form $CF^{j-1}L$: the same three matrices determine every lag. Separate regression discards these algebraic relations. It can require a long history and many coefficients when important information persists. An estimated predictor must also be checked as a generator; the binary example showed why one-step fitting alone can overlook error persistence.

This motivates a two-stage identification method: fit observable prediction coefficients, arrange them in a matrix whose low rank expresses shared dynamics, and factor that matrix to recover a recurrent predictor. The lecture sketched this route. The following supplementary results make precise what the autoregression retains and what the factorization recovers.

4.7.1 Exact finite autoregression: Cayley–Hamilton*

In a noiseless LDS, finite-dimensional hidden dynamics can be eliminated exactly. The key linear-algebra result deserves an explicit statement.

Theorem 4.7.2 (Cayley–Hamilton). *Let $A \in \mathbb{R}^{d \times d}$ have characteristic polynomial*

$$\chi_A(\lambda) = \det(\lambda I - A) = \lambda^d + a_{d-1}\lambda^{d-1} + \cdots + a_1\lambda + a_0.$$

Then

$$\chi_A(A) = A^d + a_{d-1}A^{d-1} + \cdots + a_1A + a_0I = 0. \quad (4.64)$$

The proof by the adjugate identity is in [Appendix 4.F.1](#).

Corollary 4.7.3 (Eliminating a noiseless hidden state). *If $S_{t+1} = AS_t$ and $Z_t = CS_t$, then*

$$Z_{t+d} + a_{d-1}Z_{t+d-1} + \cdots + a_0Z_t = 0. \quad (4.65)$$

Proof. The left side is $C\chi_A(A)S_t = 0$. $\square$

Even dynamics that retain information indefinitely have this recurrence. No forgetting assumption was used. With noise, the same elimination produces additional, generally correlated noise terms. To fit a predictor by regression, we instead need its conditional mean given the *observations*. That is the next construction.

4.7.2 Noisy prediction and truncation error*

For a Gaussian LDS, the conditional distribution can be computed by a recursive predictor. Its state is an estimate obtained from the observations. This gives an observable autoregression after unrolling. The useful approximation quantity is the tail of its history coefficients.

The observations in this subsection are real vectors. For their laws, log loss uses a probability density, and KL is

$$D(P, Q) = \int p(z) \log \frac{p(z)}{q(z)} dz,$$

where p, q are the densities of P, Q . The probability and KL chain rules apply to densities in the same way.

Theorem 4.7.4 (A constant-gain Gaussian predictor). *Consider Eq. (4.60) with spectral radius $\rho(A) < 1$, independent centered Gaussian process and observation noises with covariances $\Sigma_W \succ 0$ and $\Sigma_V \succ 0$. There are matrices $\Pi \succ 0$ and L such that, when $S_1 \sim \mathcal{N}(0, \Pi)$ independently of the noises, the recursion*

$$M_1 = 0, \quad M_{t+1} = FM_t + LZ_t, \quad F = A - LC \quad (4.66)$$

satisfies

$$M_t = \mathbb{E}[S_t \mid Z_{<t}], \quad Z_t \mid Z_{<t} \sim \mathcal{N}(CM_t, \Omega), \quad \Omega = C\Pi C^\top + \Sigma_V. \quad (4.67)$$

Moreover, $\rho(F) < 1$ and the observations have uniformly bounded second moments.

The recursion is the *Kalman predictor*. The covariance Π is chosen to make its gain constant; [Appendix 4.F.2](#) constructs it and proves the theorem. For an arbitrary Gaussian initialization, the same conditional calculation gives time-dependent gains. Positive noise covariance means positive variance in every nonzero direction. A stable A permits partially observed states and arbitrary C in this result.

Define the *innovations* $E_t = Z_t - CM_t$. Their conditional law is always $\mathcal{N}(0, \Omega)$, so they are independent across time and independent of the previous observations. Consequently, the same observation law can be generated by

$$Z_t = CM_t + E_t, \quad M_{t+1} = (F + LC)M_t + LE_t = AM_t + LE_t. \quad (4.68)$$

The conditional predictor and its innovations law together specify generation.

Unrolling [Eq. \(4.66\)](#) gives

$$CM_t = \sum_{j=1}^{t-1} K_j Z_{t-j}, \quad K_j = CF^{j-1}L. \quad (4.69)$$

Because F is stable, these coefficients decay geometrically. The following proposition translates their tail into a sequence-law guarantee.

Proposition 4.7.5 (Truncation controls prediction and sequence KL). *In the setting above, suppose $\|K_j\| \leq G\gamma^{j-1}$ for $j \geq 1$, with $0 < \gamma < 1$, and $\sup_t (\mathbb{E}[\|Z_t\|_2^2])^{1/2} \leq B_Z$. Define*

$$\widehat{Z}_t^{(p)} = \sum_{j=1}^{\min(p, t-1)} K_j Z_{t-j}.$$

Then

$$(\mathbb{E}[\|CM_t - \widehat{Z}_t^{(p)}\|_2^2])^{1/2} \leq a_p := \frac{GB_Z\gamma^p}{1-\gamma}. \quad (4.70)$$

Let $P_{1:T}$ be the true observation law and let $Q_{1:T}^{(p)}$ use conditional law $\mathcal{N}(\widehat{Z}_t^{(p)}, \Omega)$. Then

$$D(P_{1:T}, Q_{1:T}^{(p)}) \leq \frac{Ta_p^2}{2\lambda_{\min}(\Omega)}. \quad (4.71)$$

Proof. The triangle inequality in L^2 bounds the omitted tail by $B_Z \sum_{j>p} G\gamma^{j-1} = a_p$. For Gaussians with the same covariance,

$$D(\mathcal{N}(u, \Omega), \mathcal{N}(v, \Omega)) = \frac{1}{2}(u - v)^\top \Omega^{-1}(u - v).$$

Apply this at each observed prefix and use the KL chain rule. $\square$

This explains why an observable autoregression can approximate a noisy state-space model well: the lost remote history has an explicit cost. The displayed coefficients are the ideal comparison for fitting. Regression contributes coefficient-estimation error in addition to this truncation error. Squared loss is convenient for the convex fit and the Gaussian calculation; the state-elimination identity itself is independent of the fitting loss.

4.7.3 Recover a compact predictor by matrix factorization*

The coefficients in Eq. (4.69) come from a small recurrent computation. Their matrix structure reveals that computation. Suppose $K_j = C_\star F_\star^{j-1} L_\star$ has minimal state dimension $d_\star$. Choose a block count q for which

$$\mathcal{O}_\star = \begin{pmatrix} C_\star \\ C_\star F_\star \\ \vdots \\ C_\star F_\star^{q-1} \end{pmatrix}, \quad \mathcal{R}_\star = (L_\star, F_\star L_\star, \dots, F_\star^{q-1} L_\star)$$

both have rank $d_\star$. The first maps a state to successive readouts; the second collects how successive observations enter the state. Form the $qm \times qm$ block matrices

$$\mathcal{H}_0 = [K_{i+j-1}]_{i,j=1}^q = \mathcal{O}_\star \mathcal{R}_\star, \quad \mathcal{H}_1 = [K_{i+j}]_{i,j=1}^q = \mathcal{O}_\star F_\star \mathcal{R}_\star. \quad (4.72)$$

Each block is $m \times m$; block rows and columns index successive lags. A matrix constant along block anti-diagonals is a *block Hankel matrix*. Here $\text{rank}(\mathcal{H}_0) = d_\star$.

Take its rank- $d_\star$ singular value decomposition and factor it:

$$\mathcal{H}_0 = UDV^\top = \mathcal{O}\mathcal{R}, \quad \mathcal{O} = UD^{1/2}, \quad \mathcal{R} = D^{1/2}V^\top.$$

With † denoting the Moore–Penrose pseudoinverse, set

$$\overline{F} = \mathcal{O}^\dagger \mathcal{H}_1 \mathcal{R}^\dagger, \quad \overline{C} = \mathcal{O}_{1:m,:}, \quad \overline{L} = \mathcal{R}_{:,1:m}. \quad (4.73)$$

Proposition 4.7.6 (Recovery of the prediction map). *The recovered matrices satisfy*

$$\overline{C} \overline{F}^{j-1} \overline{L} = K_j \quad \text{for every } j \geq 1.$$

Thus the recursion

$$\overline{M}_1 = 0, \quad \widehat{Z}_t = \overline{C} \overline{M}_t, \quad \overline{M}_{t+1} = \overline{F} \overline{M}_t + \overline{L} Z_t \quad (4.74)$$

computes the same predictor on every observed history.

The proof in [Appendix 4.F.3](#) shows that the recovered states differ from minimal predictive coordinates by an invertible change of basis. Their role is to implement prediction; recovering the original physical coordinates is optional.

With estimated coefficients, form estimated Hankel matrices, retain a chosen number of singular values, and apply [Eq. \(4.73\)](#). Run [Eq. \(4.74\)](#) on data to obtain residuals. In a Gaussian innovations model, estimate their covariance and generate by

$$\tilde{Z}_t = \bar{C} \tilde{M}_t + \tilde{E}_t, \quad \tilde{M}_{t+1} = (\bar{F} + \bar{L}\bar{C})\tilde{M}_t + \bar{L}\tilde{E}_t. \quad (4.75)$$

This is what follows factorization: a predictor, a fitted innovations law, and a generator. Held-out likelihood evaluates its distributional fit; generated costs and other target statistics evaluate its intended use. Rank can be selected using these evaluations.

The two relevant update matrices have different jobs. $\bar{F}$ controls decay of observation-history coefficients. The matrix $\bar{F} + \bar{L}\bar{C}$ propagates the state during generation. Its stability margin determines whether errors in the fitted generator can persist or grow. Retaining a compact state preserves remote history without truncation; long-run accuracy still requires control of the generated dynamics.

4.7.4 Stable autoregression can also control long-run error*

Stability can be imposed directly on the observable regression. Here is one sufficient condition that preserves convex fitting.

Proposition 4.7.7 (A stable autoregressive fit). *Let $Z_{t+1} = P_t + E_{t+1}$, where P_t is its conditional mean and the E_t are iid additive innovations independent of the past. Fix fitted coefficient matrices satisfying*

$$\sum_{j=1}^p \|\hat{B}_j\| \leq \beta < 1, \quad \sup_{t \geq p} \left(\mathbb{E} \left[\left\| P_t - \sum_{j=1}^p \hat{B}_j Z_{t+1-j} \right\|_2^2 \right] \right)^{1/2} \leq \varepsilon.$$

Start a fitted generator from the same first p observations and use the same innovations in a coupling:

$$\tilde{Z}_{t+1} = \sum_{j=1}^p \hat{B}_j \tilde{Z}_{t+1-j} + E_{t+1}.$$

Then

$$\sup_t (\mathbb{E}[\|Z_t - \tilde{Z}_t\|_2^2])^{1/2} \leq \frac{\varepsilon}{1 - \beta}. \quad (4.76)$$

Proof. Let d_t denote the RMS discrepancy. Inserting the fitted predictor evaluated on the true history and using the L^2 triangle inequality gives $d_{t+1} \leq \varepsilon + \sum_j \|\hat{B}_j\| d_{t+1-j}$. Starting from $d_1 = \dots = d_p = 0$, induction gives the result. $\square$

The constraint on the sum of operator norms is convex. An L_c -Lipschitz statistic of one observation, or its time average, then has expectation discrepancy at most $L_c \varepsilon / (1 - \beta)$. [Appendix 4.F.4](#) gives the corresponding state-space comparison and a second-moment bound. Thus both state-space modeling and autoregression can use stability. The choice concerns how useful memory is represented, how much is truncated, and what constraints the fitting method preserves.

4.8 Take-home messages

1. Next-token log-loss minimization is sequence maximum likelihood. Its population excess loss is forward sequence KL.
2. Successful demonstrations specify a distribution to imitate. The task objective is the expected cost of generated sequences. TV and KL connect these objectives, with quantitative losses that matter.
3. At the same forward KL κ and TV, the laws Q_- and Q_+ have expected counts $1 - e^{-\kappa}$ and $T(1 - e^{-\kappa})$, and terminal error probabilities zero and $1 - e^{-\kappa}$. Approximate likelihood fitting can leave this choice unresolved: a per-token tolerance η permits per-sequence excess $T\eta$.
4. Longer training sequences provide more information, while longer predictions may require learning additional structure. In the grouped Bernoulli model, $d(T)$ unknown shared probabilities give expected sequence KL at most $d(T)/n$. The persistent law Q_+ shows that a vanishing average-loss gap can coexist with unchanged task error as the horizon grows.
5. Recovery attenuates an initial error; continued prediction errors contribute according to how long their effects persist. Contraction bounds both terms. Useful task memory may still need to persist.
6. Forward KL penalizes missing data-supported behavior; reverse KL penalizes implausible generated behavior. Reverse KL controls a cost according to its concentration under the data law. Obtaining this control requires more than evaluating model likelihood on supplied examples.
7. Multi-step objectives integrate over missing intermediate tokens. Direct cost optimization instead uses evaluations on model generations. Both change the computational or informational requirements.
8. State-space models express what should be remembered and how it evolves. Latent likelihoods can be difficult to fit; observed regression provides a computationally simpler route.
9. In the additional LDS results, Cayley–Hamilton gives an exact noiseless recurrence, coefficient decay bounds noisy truncation error, and Hankel factorization recovers a compact recurrent predictor. Stability constraints concern the dynamics actually used for generation.

4.9 Exercises

These exercises follow the lecture from KL and task cost to sequence length, fitting, recovery, and hidden-state models.

General instructions. Justify every answer, including yes/no answers, even when the question does not explicitly ask for a justification.

1. **The KL chain rule.** Let P, Q be laws on $[N]^T$, with $Q(x) > 0$ whenever $P(x) > 0$.
 - (a) Use the probability chain rule to write $\log(P(X_{1:T})/Q(X_{1:T}))$ as a sum.
 - (b) Take expectation under P , and then condition on $X_{<t}$ in each term, to prove the KL chain rule in Eq. (4.5).
 - (c) State the reversed identity, with its own support condition. Which prefix distribution appears in each direction?

- (d) Let P, Q now be laws on $\mathcal{U} \times [N]^T$, where $\mathcal{U}$ is a finite prompt set, with the same prompt marginal μ . Prove

$$D(P, Q) = \sum_{u: \mu(u) > 0} \mu(u) D(P(\cdot | u), Q(\cdot | u)).$$

2. **Zero-cost data and forward KL.** Let P, Q be laws on a finite set $\mathcal{X}$, and let $c : \mathcal{X} \rightarrow [0, C_{\max}]$, with $C_{\max} > 0$ and $\mathbb{E}_P[c] = 0$.

- (a) Prove the bound in [Eq. \(4.11\)](#).
 (b) Let $\kappa \geq 0$ and let R be supported on $\{x : c(x) = C_{\max}\}$. Show that $Q = e^{-\kappa}P + (1 - e^{-\kappa})R$ attains the bound.
 (c) Can a smaller bound hold for all such laws and costs using only $C_{\max}$ and $D(P, Q)$? Explain using part (b).

HINTS For (a), take $A = \{x : c(x) = 0\}$ and compare $P(A)$ with $Q(A)$.

3. **Concentration and reverse KL.** Use the notation of [Section 4.2.3](#).

- (a) Prove [Proposition 4.2.6](#) by exponential tilting, and then deduce [Theorem 4.2.8](#), including the case $v = 0$.
 (b) Let $c = T^{-1} \sum_{t=1}^T f_t(X_t)$, with independent $[0, 1]$ -valued summands under P . Prove $v_P(c) \leq 1/(4T)$, and deduce, for every law Q ,

$$|\mathbb{E}_Q[c] - \mathbb{E}_P[c]| \leq \sqrt{\frac{D(Q, P)}{2T}}. \quad (4.77)$$

What growth of reverse KL still permits vanishing discrepancy?

- (c) Let $P_\delta = \text{Bernoulli}(\delta)^{\otimes T}$, $0 < \delta < 1$, and set $m = \mathbb{E}_Q[c_{\text{cnt}}]/T$ for the error-count cost. Prove

$$D(Q, P_\delta) = T \text{kl}(m, \delta) + D(Q, P_m),$$

where $P_m = \text{Bernoulli}(m)^{\otimes T}$ and $\text{kl}(a, b) = a \log(a/b) + (1 - a) \log((1 - a)/(1 - b))$. Treat $m = 0, 1$ as well.

- (d) Which generator has the smallest reverse KL among laws with a specified expected count?

HINTS For (a), normalize $P(x)e^{\lambda(c(x) - \mathbb{E}_P[c])}$ to obtain a probability law, then use nonnegativity of KL. For (b), apply Hoeffding's lemma to each independent summand.

4. **Comparing fitting and task discrepancies.*** Use the definitions in [Section 4.2.5](#). For parts (b)–(e), let $P = \delta_{0^T}$, $T \geq 2$, with $c_{\text{cnt}}(x) = \sum_{t=1}^T x_t$ and $c_{\text{fin}}(x) = x_T$.

- (a) Prove that d_c is a pseudometric. Give distinct laws with $d_c(P, Q) = 0$, first for the binary count and then for another cost.
 (b) Use the fixed-KL intervals in [Eqs. \(4.24\)](#) and [\(4.25\)](#) to prove

$$\omega_{D \rightarrow c_{\text{cnt}}}(\kappa; P) = T(1 - e^{-\kappa}), \quad (4.78)$$

$$\omega_{D \rightarrow c_{\text{fin}}}(\kappa; P) = 1 - e^{-\kappa}, \quad \kappa \geq 0. \quad (4.79)$$

Which of Q_- and Q_+ attains these bounds?

(c) Calculate the reverse conversions:

$$\omega_{c_{\text{cnt}} \rightarrow D}(s; P) = \begin{cases} -\log(1-s), & 0 \leq s < 1, \\ +\infty, & s \geq 1, \end{cases} \quad (4.80)$$

$$\omega_{c_{\text{fin}} \rightarrow D}(s; P) = +\infty, \quad s \geq 0. \quad (4.81)$$

Exhibit laws attaining each finite bound and laws with infinite KL where appropriate. Explain why small expected count forces small KL, whereas perfect final answers allow arbitrarily large KL.

(d) Fix an expected count $s \in [0, 1)$. Prove the sharp bounds

$$-\log(1-s/T) \leq D(P, Q) \leq -\log(1-s).$$

Give a law attaining each endpoint and relate them to part (c).

- (e) Replace the count by the average count c_{cnt}/T . Recompute both moduli. How does normalization change their scales and the ratio between the largest and smallest costs at a fixed positive KL?
- (f) Verify that the cost in [Proposition 4.2.9](#) may have zero variance under P and positive expectation under Q . Which term in the bound accounts for this?

5. **Repeated errors and their accumulation.** Compare isolated errors with errors that persist. Fix $T \geq 2$, $P = \delta_0^T$, $0 < \varepsilon < 1$, and $X_0 = 0$. Use $c_{\text{cnt}}(x) = \sum_{t=1}^T x_t$ and $c_{\text{fin}}(x) = x_T$. The Markov generators $Q_{\text{ind}}, Q_{\text{bad}}$ use

	$Q(X_{t+1} = 1 \mid X_t = 0)$	$Q(X_{t+1} = 1 \mid X_t = 1)$	$(0 \leq t < T).$
Q_{ind}	ε	ε	
Q_{bad}	ε	1	

For P , use the kernel that returns zero from either state.

(a) Show that both generators have conditional KL $-\log(1-\varepsilon)$ at every data prefix, and hence

$$D(P, Q_{\text{ind}}) = D(P, Q_{\text{bad}}) = T \log \frac{1}{1-\varepsilon}. \quad (4.82)$$

Compare this constant positionwise loss with the loss profile of Q_- and Q_+ . Compute the probability that the first one occurs at position s , and the probability of no ones.

(b) Prove

$$\mathbb{E}_{Q_{\text{ind}}}[c_{\text{cnt}}] = T\varepsilon, \quad (4.83)$$

$$\mathbb{E}_{Q_{\text{bad}}}[c_{\text{cnt}}] = \sum_{t=1}^T [1 - (1-\varepsilon)^t] = T - \frac{(1-\varepsilon)[1 - (1-\varepsilon)^T]}{\varepsilon}. \quad (4.84)$$

Obtain the second identity both from marginal error probabilities and by conditioning on the first error position. Also prove

$$\mathbb{E}_{Q_{\text{ind}}}[c_{\text{fin}}] = \varepsilon, \quad \mathbb{E}_{Q_{\text{bad}}}[c_{\text{fin}}] = 1 - (1-\varepsilon)^T. \quad (4.85)$$

- (c) Prove the finite bounds

$$\frac{\varepsilon T(T+1)}{2} - \frac{\varepsilon^2 T(T+1)(T-1)}{6} \leq \mathbb{E}_{Q_{\text{bad}}}[c_{\text{cnt}}] \leq \frac{\varepsilon T(T+1)}{2}.$$

Deduce $\mathbb{E}_{Q_{\text{bad}}}[c_{\text{cnt}}] \sim \varepsilon T(T+1)/2$ when $T\varepsilon \rightarrow 0$. Explain why the expected count can grow quadratically with T in this regime.

- (d) Show that both sequence TVs equal
- $1 - (1 - \varepsilon)^T$
- . For either generator, verify the bounds

$$\mathbb{E}_Q[c_{\text{cnt}}] \leq T[1 - (1 - \varepsilon)^T], \quad \text{using exact TV,} \quad (4.86)$$

$$\mathbb{E}_Q[c_{\text{cnt}}] \leq T \min \left\{ 1, \sqrt{\frac{T}{2} \log \frac{1}{1 - \varepsilon}} \right\}, \quad \text{using Pinsker.} \quad (4.87)$$

Compare these with the exact counts at $T = 1000, \varepsilon = 10^{-3}$, and their leading terms when $T\varepsilon \rightarrow 0$. Which information about error duration does the TV-to-count bound discard?

- (e) Set
- $\varepsilon = 1 - e^{-\kappa/T}$
- for fixed
- $\kappa > 0$
- . Compute the two terminal costs and the ratio
- $\mathbb{E}_{Q_{\text{bad}}}[c_{\text{fin}}]/\mathbb{E}_{Q_{\text{ind}}}[c_{\text{fin}}]$
- . Explain why this ratio is of order
- T
- even though the terminal cost range is fixed. Which model attains the terminal-cost upper endpoint?

6. **Positionwise and average KL bounds.** Let $P = \delta_{0^T}$, $T \geq 2$, and use the counting and terminal costs from [Exercise 5](#). Compare a bound on each next-token KL with a bound on their average. Fix $\eta \geq 0$.

- (a) Suppose a generator
- Q
- satisfies

$$\mathbb{E}[D(P(\cdot | X_{<t}), Q(\cdot | X_{<t}))] \leq \eta \quad (t = 1, \dots, T), \quad X_{1:T} \sim P.$$

Show that $Q(X_{1:t} = 0^t) \geq e^{-t\eta}$, and deduce

$$\mathbb{E}_Q[c_{\text{cnt}}] \leq \sum_{t=1}^T (1 - e^{-t\eta}), \quad \mathbb{E}_Q[c_{\text{fin}}] \leq 1 - e^{-T\eta}. \quad (4.88)$$

Show that Q_{bad} attains both bounds when $\varepsilon = 1 - e^{-\eta}$.

- (b) Suppose instead that only
- $D(P, Q)/T \leq \eta$
- . Use
- [Eqs. \(4.24\)](#)
- and
- [\(4.25\)](#)
- to obtain the corresponding bounds. Compare their small-
- $T\eta$
- leading terms with
- [Eq. \(4.88\)](#)
- . What per-token accuracy suffices to guarantee terminal error at most
- $s \in (0, 1)$
- ?

7. **Estimating shared parameters.*** Use the independent-bit models and pooled estimator from [Section 4.4.1](#). All expectations below average over the training observations. Fix $0 < a < 1/2$.

- (a) Let
- $B \sim \text{Binomial}(L, p)$
- , with
- $L \geq 1$
- . For
- $0 < p < 1$
- , show that the ordinary empirical frequency
- $\bar{p} = B/L$
- satisfies
- $\mathbb{E}[\text{kl}(p, \bar{p})] = \infty$
- . Explain why adding one count of each bit gives
- $\hat{p} = (B+1)/(L+2)$
- , and why this assigns positive probability to both bits. To obtain a finite-sample bound for this choice, prove that, for
- $0 < p < 1$
- ,

$$\mathbb{E} \left[\frac{1}{B+1} \right] = \frac{1 - (1-p)^{L+1}}{(L+1)p}.$$

Apply the same identity to $L - B$, then use Jensen's inequality for the logarithm to show

$$\mathbb{E}[\text{kl}(p, \hat{p})] \leq \log \frac{L+2}{L+1} \leq \frac{1}{L+1}.$$

Verify the bound directly for $p = 0, 1$.

- (b) Apply part (a) with $L = nm_j$ to bound $\kappa_n(T)$ from above. For the lower bound, calculate

$$\mathbb{E}[(\hat{p} - p)^2] = \frac{Lp(1-p) + (1-2p)^2}{(L+2)^2}.$$

Use Pinsker's inequality and $L/(L+2) \geq 1/3$ to show that

$$\frac{2a(1-a)}{9} \frac{d(T)}{n} \leq \kappa_n(T) \leq \frac{d(T)}{n} \quad \text{when } p_j \in [a, 1-a].$$

Explain why group sizes cancel from both bounds.

- (c) For fixed $p \in (0, 1)$, prove $L \mathbb{E}[\text{kl}(p, \hat{p})] \rightarrow 1/2$ as $L \rightarrow \infty$. Deduce Eqs. (4.28) and (4.29), keeping track of which of n, T is fixed in each limit. Make the expansion uniform for $p \in [a, 1-a]$ and prove Eq. (4.32) when $n \min_j m_j \rightarrow \infty$.

HINTS For (a), use $1/(B+1) = \int_0^1 u^B du$. For (c), expand $\text{kl}(p, q)$ near $q = p$ and use a binomial tail bound for the remaining event.

8. **Sequence length and the training budget.*** Use the grouped Bernoulli model and smoothed estimator from Section 4.4.1. Assume $p_j \in [a, 1-a]$, with $0 < a < 1/2$.

- (a) Keep n fixed and suppose $p_t \in [a, 1-a]$. Use Exercise 7(b) to give explicit upper and lower bounds proportional to T/n for the unshared model's expected sequence KL. Compare this growth with the shared model's limit in Eq. (4.28).
- (b) Use one new parameter for each successive block of lengths $1, 2, 3, \dots$. Compute $d(T)$ at complete-block endpoints and interpret Eqs. (4.31) and (4.32). How can normalized KL improve while sequence KL grows?
- (c) Fix the total training-token budget $L = nT$ and use $\kappa'_L(T) = \kappa_{L/T}(T)$ to repeat the shared and unshared leading-order comparisons. Explain why increasing T worsens expected sequence KL in the shared model. Compare the numbers of observations per parameter in the two models. Translate the expected-KL bounds into bounds on the discrepancy of a cost in $[0, 1]$.

9. **Task costs at a given learning rate.*** Prove Proposition 4.4.2 using the following construction. At each horizon T , partition the first T positions into $\lceil d(T) \rceil$ nonempty groups. Let the U_t be independent fair bits and $V_t = 0$ under P . Use terminal cost V_T and counting cost $\sum_{t=1}^T V_t$. Write P_T and P_T^U for the laws of $(U, V)_{1:T}$ and $U_{1:T}$, respectively. If $\hat{P}_T^U$ is the pooled smoothed estimate of the law of $U_{1:T}$, set

$$r_T = \mathbb{E}[D(P_T^U, \hat{P}_T^U)]$$

and, for any law R supported on $v_{1:T} \neq 0^T$, define

$$\hat{Q}_R = e^{-r_T} (\hat{P}_T^U \otimes \delta_{0^T}) + (1 - e^{-r_T}) R.$$

- (a) Show that r_T is determined by n, T and the chosen partition, so the mixture weight can be used by the estimator.
- (b) Compute $\mathbb{E}[D(P_T, \hat{Q}_R)]$ and show that it has order $d(T)/n$, independently of R .
- (c) Determine the full attainable intervals for the expected terminal and counting costs as R varies. Identify laws attaining the endpoints.

10. **Identifying one of two Bernoulli models.*** The data law is one of two infinite i.i.d. Bernoulli processes, with parameter $p \in \{1/4, 3/4\}$. Observe n independent length- T sequences. Choose $\hat{p} = 3/4$ if the empirical mean of their nT bits is at least $1/2$, and choose $1/4$ otherwise. Let $\hat{Q}$ be the i.i.d. model with parameter $\hat{p}$.

- (a) Prove that the probability of choosing the wrong parameter is at most $(\sqrt{3}/2)^{nT} \leq e^{-nT/8}$.
 (b) Compute the sequence KL when the selected parameter is wrong, and deduce

$$\kappa_n(T) \leq \frac{T \log 3}{2} e^{-nT/8} \rightarrow 0 \quad (T \rightarrow \infty, n \text{ fixed}).$$

- (c) Compare this with estimating an arbitrary $p \in (0, 1)$ in Eq. (4.28). What allows the two-candidate learner to identify the entire model exactly?

HINTS For (a), let B be the number of observed ones and apply Markov's inequality to $3^{B-nT/2}$ when $p = 1/4$.

11. **Copying with a bounded context.*** Let P generate $B_1 B_1 B_2 B_2 \dots$, where the blocks B_m are independent and uniform on $\{0, 1\}^m$. For an integer $w \geq 0$, let $\mathcal{Q}_w$ be all infinite-sequence models whose next-token distribution can depend on the position and the last w tokens only.

- (a) In a copied block of length $m > w$, show that the required bit is fair given the available context, although the full prefix determines it. Deduce expected conditional KL at least $\log 2$ at each such position.
 (b) At $T = M(M + 1)$, for $M > w$, prove

$$\inf_{Q \in \mathcal{Q}_w} D(P_{1:T}, Q_{1:T}) = \frac{\log 2}{2} (T - w(w + 1)). \quad (4.89)$$

Specify a predictor attaining equality.

- (c) Explain why more training data cannot remove this error for a fixed w . How must w grow to predict every copied bit exactly?
 (d) Show that an exact recurrent generator must distinguish 2^m states after generating a fresh length- m block, and hence needs at least m bits of state information. Compare the description length of the copying rule with its memory requirement. Relate the distinction to recalling earlier definitions or intermediate results in a mathematical argument.
12. **A missed dependency at the final prediction.*** Let $T \geq 2$. Under P_{par} , the first $T - 1$ bits are independent fair coins and the last is their parity: $X_T = (\sum_{t < T} X_t) \bmod 2$. Under Q_{coin} , all T bits are independent fair coins. The task cost is

$$c_{\text{par}}(x_{1:T}) = \mathbb{I} \left\{ x_T \neq \left(\sum_{t < T} x_t \right) \bmod 2 \right\}.$$

- (a) Compute the next-token KL at every position and prove

$$\begin{aligned} D(P_{\text{par}}, Q_{\text{coin}}) &= \log 2, & \frac{D(P_{\text{par}}, Q_{\text{coin}})}{T} &= \frac{\log 2}{T}, \\ \mathbb{E}_{P_{\text{par}}}[c_{\text{par}}] &= 0, & \mathbb{E}_{Q_{\text{coin}}}[c_{\text{par}}] &= \frac{1}{2}. \end{aligned} \quad (4.90)$$

Verify that Eq. (4.11) is attained. Compare the effect of normalization with Eq. (4.34).

- (b) For the model class $\{P_{\text{par}}, Q_{\text{coin}}\}$ and any sample from P_{par} , prove that returning Q_{coin} has empirical per-token optimization gap $(\log 2)/T$. What has been left unlearned when this gap is accepted?
- (c) Now use the class of all independent-bit laws $Q = \bigotimes_{t=1}^T \text{Bernoulli}(q_t)$. Show that every bit is marginally fair under P_{par} , and deduce that Q_{coin} uniquely minimizes population log loss in this class. Explain why the same prediction failure is now approximation error.
- (d) If $D(P, Q)/T \leq \eta$, what is the largest possible final-position contribution

$$\mathbb{E}[D(P(\cdot | X_{<T}), Q(\cdot | X_{<T}))], \quad X_{1:T} \sim P,$$

in Eq. (4.5)? Give an example attaining it. Compare the information supplied by the average and by the individual positionwise losses.

- (e) Define the running parity $Y_t = (\sum_{j=1}^t X_j) \bmod 2$. Describe its Markov kernels under P_{par} and Q_{coin} . Apply Theorem 4.5.2 to this state, for which $c_{\text{par}} = Y_T$. Show that the fitted kernels have contraction factor zero, all earlier local discrepancies vanish, and the final local discrepancy is $1/2$. Why does contraction leave this final error unchanged?
13. **Certifying an optimization tolerance.*** Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable, with $f_\star = \inf_{v \in \mathbb{R}^d} f(v) > -\infty$. For $f(\theta) = \hat{L}(Q_\theta)$ and $\mathcal{Q} = \{Q_\theta : \theta \in \mathbb{R}^d\}$, the gap $f(\theta) - f_\star$ is the optimization error in Eq. (4.35).

- (a) Suppose f is μ -strongly convex, meaning that for a known $\mu > 0$ and every $\theta, v \in \mathbb{R}^d$,

$$f(v) \geq f(\theta) + \nabla f(\theta)^\top (v - \theta) + \frac{\mu}{2} \|v - \theta\|_2^2.$$

Prove

$$f(\theta) - f_\star \leq \frac{\|\nabla f(\theta)\|_2^2}{2\mu}.$$

For a desired per-sequence tolerance $s > 0$, give a sufficient gradient-norm threshold that certifies $T(f(\theta) - f_\star) \leq s$.

- (b) Take $d = 1$ and $f(\theta) = (\theta^2 - 1)^2$. Compute $f'(0)$ and $f(0) - f_\star$. Explain why a small gradient alone does not certify a small optimization error for a general loss.

HINTS For (a), minimize the quadratic lower bound over $v - \theta$.

14. **Likelihood accuracy, data, and support.*** Parts (a)–(c) use the repeated-error generators from Exercise 5 and all-zero training sequences.

- (a) Compute both empirical per-token losses. Include $P = \delta_{0^T}$ in the model class and show that $\varepsilon = 1 - e^{-\eta}$ makes both generators acceptable at optimization tolerance $\eta > 0$.
- (b) Compute their terminal and counting costs and their leading terms as $T\eta \rightarrow 0$. Give a sufficient order of η in T to keep each expected count bounded.
- (c) Remove P and use a finite collection of paired generators at distinct positive values of ε . Identify all empirical minimizers. Explain why exact fitting leaves a tie between models of different task cost.
- (d) Specialize Theorem 4.4.4 to zero-cost data, first for $c \in [0, T]$ and then for $c \in [0, 1]$. Express each guarantee using the number of sequences n , then using the token budget $L = nT$.

- (e) In [Proposition 4.D.1](#), choose a prompt on which the two candidate rules disagree. Show that its expert label identifies the correct rule. Compare the number of such queries with the number of random demonstrations needed for failure probability at most $\delta \in (0, 1/2)$.
15. **Computing further-ahead predictions.** Use the autoregressive model in [Eq. \(4.44\)](#). Fix $x_{\leq t}$, $k \geq 2$ with $t + k \leq T$, and $a \in [N]$.
- (a) Verify [Eq. \(4.42\)](#). The sequence likelihoods of Q_- , Q_+ agree; which prediction question distinguishes them, and which feature of their generation does it evaluate?
 - (b) Derive [Eq. \(4.44\)](#). Count its terms and the number of conditional evaluations if each term is evaluated separately. Explain why sharing prefixes still leaves exponentially many distinct prefixes in a general model.
 - (c) Explain which conditional probability is evaluated if all the intervening tokens are replaced by their observed values.
 - (d) Suppose instead that the model is first-order Markov on $[N]$ with known transition matrix K . Give a recursion that computes the entire k -step distribution in $O(kN^2)$ arithmetic operations. Why can this recursion merge histories?
 - (e) A separate head $h_{\theta,k}(\cdot | x_{\leq t})$ can be trained from pairs $(X_{\leq t}, X_{t+k})$. State the additional equality needed for its probabilities to be the original sequence model's k -step marginals.
16. **Contraction and repeated errors.** This exercise compares recovery from an initial error with continual correction of new errors.
- (a) Prove [Eq. \(4.53\)](#).
 - (b) For the recovery family Q_r , compute the contraction factor and use [Eq. \(4.51\)](#) to recover the terminal and count bounds in [Proposition 4.5.3](#). Verify that the recurrence inequality is an equality here, giving the exact costs.
 - (c) Now let $Q_{\varepsilon,r}$, with $X_0 = 0$, use

$$Q_{\varepsilon,r}(X_{t+1} = 1 | X_t = 0) = \varepsilon, \quad Q_{\varepsilon,r}(X_{t+1} = 0 | X_t = 1) = r,$$

where $0 < \varepsilon < 1$, $0 \leq r \leq 1$, and $0 \leq t < T$. For $p_t = Q_{\varepsilon,r}(X_t = 1)$, prove

$$p_0 = 0, \quad p_{t+1} = \varepsilon + (1 - \varepsilon - r)p_t, \quad p_t = \frac{\varepsilon}{\varepsilon + r}[1 - (1 - \varepsilon - r)^t]. \quad (4.91)$$

Identify the choices of r giving the two models in [Exercise 5](#).

- (d) Compute the contraction factor $|1 - \varepsilon - r|$. For $0 \leq r \leq 1 - \varepsilon$, prove

$$\mathbb{E}_{Q_{\varepsilon,r}}[c_{\text{fin}}] \leq \frac{\varepsilon}{\varepsilon + r}, \quad \frac{1}{T} \mathbb{E}_{Q_{\varepsilon,r}}[c_{\text{cnt}}] \leq \frac{\varepsilon}{\varepsilon + r}. \quad (4.92)$$

Recover these bounds from [Theorem 4.5.2](#), using $e_1 = \varepsilon$ and local error ε at every subsequent step. At $r = 0$, why does the positive contraction margin still give the bound one? How large must r be relative to ε for the limiting error probability to be at most $s \in (0, 1)$?

HINTS For (a), normalize the positive and negative parts of $\alpha - \beta$ to obtain two probability laws.

17. **Forgetting errors and retaining information.** Use the reset mixture and persistent-state model from [Section 4.5.3](#).
- (a) Prove the contraction bound for the reset mixture K_r . Construct a process whose next-token law uses only the last token and whose initial bit can be recovered from every later token.

- (b) In Eq. (4.56), suppose the conditional working-state kernels contract uniformly with factor $\rho < 1$. Explain why contraction applies when comparing equal M , while different persistent M 's can remain distinguishable indefinitely.
18. **Latent likelihood and the forward algorithm.** Use the hidden Markov model in Eq. (4.58), with d hidden states, transition matrix A , emission matrix C , and initial law π .

- (a) Derive the likelihood by summing the joint law of hidden states and observations. How many hidden trajectories appear?
- (b) Starting from $\Lambda_0 = \pi$, derive the recursion

$$\Lambda_t = \text{diag}(C_{X_{t,1}}, \dots, C_{X_{t,d}}) A \Lambda_{t-1}, \quad Q_\theta(X_{1:T}) = \mathbf{1}^\top \Lambda_T.$$

Prove that $\Lambda_t(j)$ is the joint mass the model assigns to the observed prefix and hidden state j at time t . Explain the $O(Td^2)$ arithmetic cost.

- (c) If $J_{0:T}$ were observed, write the transition and emission counts that appear in the complete-data log likelihood. What replaces those counts when the hidden trajectory is averaged under its posterior given $X_{1:T}$?
- (d) For a continuous state model with initial density π_θ , transition density $a_\theta(s' | s)$, and emission density $b_\theta(x | s)$, write the corresponding trajectory integral.
19. **Observable recurrence and predictive state.*** Use $M_{t+1} = FM_t + LZ_t$, $\hat{Z}_t = CM_t$, and $K_j = CF^{j-1}L$ from Eqs. (4.66) and (4.69).

- (a) Let $S_{t+1} = AS_t$, where $A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$, and let $Z_t = c^\top S_t$. Use Cayley–Hamilton to derive an order-two recurrence for Z_t . Show that distinct initial states remain at the same Euclidean distance under these dynamics.
- (b) Unroll Eq. (4.66) from a nonzero fixed M_1 . Which initialization term must be added to Eq. (4.69)?
- (c) Verify the two block factorizations in Eq. (4.72), including the index shifts.
- (d) Show directly that replacing (F, C, L) by $(T_0^{-1}FT_0, CT_0, T_0^{-1}L)$, with T_0 invertible, preserves every history coefficient.
- (e) For scalars $F = 1/2, C = 1, L = 3/4$, compare predictor-memory decay with the dynamics used for generation. Which matrix is relevant to each question?

Glossary

Sequence model; autoregressive factorization

A joint law on token sequences, expressed as a product of next-token conditionals. Every finite sequence law has such a factorization.

Maximum likelihood; teacher forcing

Maximum likelihood minimizes empirical negative log probability. Teacher forcing evaluates each conditional using the observed prefix. Generation instead feeds sampled tokens back into the predictor.

Positive-data fine-tuning

Continuing model fitting on examples selected to demonstrate desired behavior. The idealization here uses sequences of cost zero.

Cost and reward

A numerical assessment of a generated response or sequence. The task objective is its expected cost, or expected reward.

Cost discrepancy; moduli of continuity

The pseudometric $d_c(P, Q) = |\mathbb{E}_P[c] - \mathbb{E}_Q[c]|$ compares one expected cost. The modulus $\omega_{D \rightarrow c}$ asks how much cost discrepancy a KL budget permits. The converse $\omega_{c \rightarrow D}$ asks how large KL can be at a given cost-discrepancy budget. The upper and lower endpoints of the fixed-KL cost ranges determine these two moduli. Taking the supremum over reference laws gives a uniform conversion.

Hellinger distance (appendix)

A distributional distance used in the proofs of the variance-sensitive cost and finite-sample likelihood bounds; defined in [Appendix 4.C.1](#).

Optimization error and tolerance

The error $E(\hat{Q})$ is the model's excess empirical per-token loss above the infimum in the class. A tolerance η bounds this error and permits per-sequence excess $T\eta$. The average can place this excess unevenly across positions.

Total variation

$\|P - Q\|_{\text{TV}}$ is the largest discrepancy in event probabilities. For costs in $[0, C_{\max}]$, it bounds expectation discrepancy multiplied by $C_{\max}$.

Forward and reverse KL

$D(P, Q)$ averages log density ratios under the data law; $D(Q, P)$ averages them under the model law. The sequence chain rules inherit these different prefix distributions.

Markov process; transition kernel

A process whose current state determines the conditional next-state law. $K(x, y)$ gives its transition probabilities.

Contraction margin

If a kernel contracts TV by at most $\rho < 1$, its margin is $1 - \rho$. The local-error bound is amplified by its reciprocal.

Recovery

In the binary example, the probability of returning from an erroneous state to zero. It controls the duration of an erroneous run.

Subgaussian cost

A cost whose centered log exponential moment is at most $\lambda^2 v^2/2$. The quantity v^2 is a variance proxy; $v_P(c)$ is the smallest valid proxy under P . It satisfies $\sigma_P(c)^2 \leq v_P(c) \leq C_{\max}^2/4$ for $0 \leq c \leq C_{\max}$.

State-space model; latent state

A model of observations through an evolving, possibly unobserved state. State coordinates specify the information retained over time.

One-hot encoding; martingale difference

$Z_t = e_{X_t}$ turns token probabilities into vector conditional means. Subtracting a conditional mean produces a residual with conditional expectation zero given the past.

Linear dynamical system; linear autoregression

An LDS has linear state and observation equations. A linear autoregression predicts from a linear combination of observed lags.

System identification

Estimating dynamical and observation equations from observed data.

Innovation; Kalman predictor

An innovation is the observation minus its conditional-mean prediction. For the Gaussian LDS considered here, the Kalman predictor computes that mean recursively, and innovations are independent Gaussian vectors.

Truncation error

The predictive effect of history omitted by a finite-lag approximation. It is controlled by the tail of the history coefficients.

Hankel factorization

Factoring a matrix of lag coefficients to recover a finite-dimensional recurrent predictor. Its rank expresses the minimal predictive state dimension under the stated rank assumptions.

Variational approximation; ELBO

Optimizing over a tractable family to approximate a target distribution. The evidence lower bound approximates latent-model log likelihood; its gap is a reverse KL between a trial and exact hidden-state posterior.

4.A Where finiteness is used

The finite sequence space guarantees bounded costs and permits term-by-term differentiation. This section records which properties the final set of results uses, so we can also apply them to other probability spaces. The remaining appendices follow the order in which their topics first require deferred arguments in the main text: reverse-KL concentration, forward-KL variance control, finite-sample fitting, variational objectives, and LDS identification.

Distributional comparisons. Data processing, the TV bound ([Proposition 4.2.2](#)), Pinsker’s inequality, the zero-cost bound ([Proposition 4.2.4](#)), the exponential-moment bound ([Proposition 4.2.6](#)), and the variance-sensitive bounds ([Propositions 4.2.9](#) and [4.C.1](#)) hold on general probability spaces with the same bounded-cost assumptions. Their proofs use integrals of densities in place of sums. The mixture and conditioning identities in [Proposition 4.5.1](#) also extend, with integrable costs. The support requirement becomes *absolute continuity*: finite $D(Q, P)$ requires $P(A) = 0 \Rightarrow Q(A) = 0$ for every measurable event A . The Markov stability argument in [Theorem 4.5.2](#) uses only the stated contraction and local-error bounds; replace the local-error sum in [Eq. \(4.50\)](#) by an expectation.

Extrema at fixed KL. For fixed P , each nonempty KL level set in [Eq. \(4.17\)](#) is compact. Indeed, the probability simplex is compact, and $D(P, Q)$ is continuous wherever $Q(x) > 0$ on the support of P . It tends to infinity if one of those probabilities tends to zero. Hence every finite level set is closed. The infinite level set is the finite union of the closed faces $\{Q : Q(x) = 0\}$ with $P(x) > 0$. Since $d_c(P, Q)$ is continuous in Q , its minimum and maximum on each level are attained. This justifies both endpoint formulas in [Proposition 4.2.10](#). On other spaces, compact level sets and continuity of the cost discrepancy give the same conclusion. The budget definitions of the moduli, their monotonicity, and the lower-bound inversion implication use only well-defined discrepancies; identifying an exact converse modulus from lower endpoints additionally uses attainment of those endpoints.

Boundedness and exponential moments. Every real-valued cost on $[N]^T$ is bounded, so its exponential moments exist for every λ and it has a finite subgaussian proxy. Bounded costs on any probability space have these same properties. For an unbounded cost, [Proposition 4.2.6](#) still holds when $\mathbb{E}_P[|c|], \mathbb{E}_Q[|c|] < \infty$, using those $\lambda > 0$ for which $\mathbb{E}_P[e^{\lambda c}] < \infty$. The subgaussian condition in [Theorem 4.2.8](#) supplies the needed exponential moments in both directions. Attainment of the optimal proxy requires only that a finite valid proxy exist; [Appendix 4.B.2](#) proves this directly. The entropy decomposition in [Eq. \(4.5\)](#) uses finite $H(P)$; for countable alphabets, $H(P) < \infty$ keeps that decomposition meaningful for comparing log loss with KL.

Differentiating expectations. Finite sums justify the variational gradient in [Appendix 4.E](#). For general densities, sufficient conditions are differentiability on a fixed support and integrable bounds, uniform in a neighborhood of the parameter, on the derivatives of the density and the objective integrand. Dominated convergence then permits differentiation under the integral. For the exponential tilts in [Appendix 4.B](#), boundedness of c supplies such bounds on every bounded interval of λ .

The maximum-cost endpoint in the exact modulus. The proof of [Proposition 4.B.1](#) uses finiteness to obtain a maximum cost m with $P(c = m) > 0$. For a bounded cost on a general space, take m to be its *essential supremum* under P : the smallest number with $P(c \leq m) = 1$. If

$P(c = m) > 0$, the same endpoint argument applies. If $P(c = m) = 0$, the tilted KL increases to infinity: the tilted laws concentrate on sets $\{c > m - \varepsilon\}$ whose P -probabilities decrease to zero. Every finite positive KL budget is therefore reached by a finite tilt. Thus the exact-modulus identity still holds for bounded costs; finiteness simplifies its endpoint proof.

Finite model classes and finite-dimensional representations. These are separate assumptions. The finite-class guarantee in [Theorem 4.4.4](#) uses a union bound over models, producing $\log |\mathcal{Q}|$; its likelihood-ratio argument also works for general observation spaces with densities. The one-hot construction in [Proposition 4.6.1](#) uses the vocabulary size as its vector dimension, and [Theorem 4.7.2](#) uses the finite dimension of a matrix. These finiteness assumptions enter the constructions and bounds themselves.

4.B Reverse KL and exponential tilting

4.B.1 From exponential moments to subgaussian control

Proposition 4.2.6 (Cost control from exponential moments). *Let P, Q be distributions over a finite set $\mathcal{X}$ and let $c : \mathcal{X} \rightarrow \mathbb{R}$. Then.*

$$\mathbb{E}_Q[c] - \mathbb{E}_P[c] \leq \inf_{\lambda > 0} \frac{D(Q, P) + \log \mathbb{E}_P[e^{\lambda(c - \mathbb{E}_P[c])}]}{\lambda}. \quad (4.12)$$

Proof. The basic inequality uses the full exponential-moment function of the cost. Put $\psi(\lambda) = \log \mathbb{E}_P[e^{\lambda(c - \mathbb{E}_P[c])}]$. Define the tilted law by

$$P_\lambda(x) = P(x)e^{\lambda(c(x) - \mathbb{E}_P[c]) - \psi(\lambda)}.$$

This is a probability law. Writing $\kappa = D(Q, P)$, we have

$$0 \leq D(Q, P_\lambda) = \kappa - \lambda(\mathbb{E}_Q[c] - \mathbb{E}_P[c]) + \psi(\lambda).$$

Consequently,

$$\mathbb{E}_Q[c] - \mathbb{E}_P[c] \leq \inf_{\lambda > 0} \frac{\kappa + \psi(\lambda)}{\lambda},$$

which proves [Proposition 4.2.6](#). □

Under the subgaussian condition,

$$\mathbb{E}_Q[c] - \mathbb{E}_P[c] \leq \inf_{\lambda > 0} \left(\frac{\kappa}{\lambda} + \frac{\lambda v^2}{2} \right) = v\sqrt{2\kappa}.$$

Apply the same argument to $-c$ for the other direction. If $v = 0$, take $\lambda \rightarrow \infty$ in each direction.

For completeness, the exponential-moment bound for averages follows from Hoeffding's lemma. If $Y \in [0, 1]$, let $h(s) = \log \mathbb{E}[e^{s(Y - \mathbb{E}[Y])}]$. Then $h(0) = h'(0) = 0$, and $h''(s)$ is the variance of Y under its tilted law. Every law on $[0, 1]$ has variance at most $1/4$, giving $h(s) \leq s^2/8$. For independent $f_t(X_t) \in [0, 1]$,

$$\log \mathbb{E} \exp \left[\frac{\lambda}{T} \sum_{t=1}^T (f_t(X_t) - \mathbb{E}[f_t(X_t)]) \right] \leq T \frac{(\lambda/T)^2}{8} = \frac{\lambda^2}{8T}.$$

Hence the variance proxy is $v^2 = 1/(4T)$.

4.B.2 Optimal variance proxies

The smallest valid proxy is attained whenever any finite proxy exists, even on an infinite probability space. To see this, let c be an integrable subgaussian cost and write $\psi(\lambda) = \log \mathbb{E}_P[e^{\lambda(c - \mathbb{E}_P[c])}]$. The constraints defining a valid proxy give the identity

$$v_P(c) = \sup_{\lambda \neq 0} \frac{2\psi(\lambda)}{\lambda^2}.$$

This value satisfies every constraint, so the set of valid proxies is exactly $[v_P(c), \infty)$.

For the bounded costs used here, expansion at zero gives

$$\psi(\lambda) = \frac{\lambda^2 \sigma_P(c)^2}{2} + o(\lambda^2),$$

and hence $\sigma_P(c)^2 \leq v_P(c)$. Scaling Hoeffding's lemma from $[0, 1]$ to $[0, C_{\max}]$ gives $\psi(\lambda) \leq \lambda^2 C_{\max}^2/8$, so $v_P(c) \leq C_{\max}^2/4$.

4.B.3 The exact reverse-KL cost modulus

How much can the expected cost increase if the reverse KL is at most κ ? Retaining the full exponential-moment function answers this question exactly when we allow all model laws.

Proposition 4.B.1 (Exact reverse-KL cost conversion). *Fix a law P and a cost c . Let $\psi(\lambda) = \log \mathbb{E}_P[e^{\lambda(c - \mathbb{E}_P[c])}]$. For every $\kappa \geq 0$,*

$$\sup_{Q: D(Q, P) \leq \kappa} (\mathbb{E}_Q[c] - \mathbb{E}_P[c]) = \inf_{\lambda > 0} \frac{\kappa + \psi(\lambda)}{\lambda}. \quad (4.93)$$

The supremum ranges over all probability laws Q .

Proof. Equation (4.12) gives the upper bound. We construct laws attaining it. Put $\mu = \mathbb{E}_P[c]$ and use the tilted laws $P_\lambda(x) = P(x)e^{\lambda(c(x) - \mu) - \psi(\lambda)}$. Differentiating ψ gives

$$\psi'(\lambda) = \mathbb{E}_{P_\lambda}[c] - \mu, \quad D(P_\lambda, P) = \lambda\psi'(\lambda) - \psi(\lambda).$$

If c is constant on $\text{supp}(P)$, both sides of Eq. (4.93) are zero. Otherwise let

$$m = \max_{x: P(x) > 0} c(x), \quad \mathcal{A} = \{x : P(x) > 0, c(x) = m\}, \quad \kappa_* = -\log P(\mathcal{A}) > 0.$$

As λ increases from zero to infinity, the tilted KL increases continuously from zero to κ_* . Indeed, its derivative is $\lambda\psi''(\lambda) > 0$, and $P_\lambda \rightarrow P(\cdot | \mathcal{A})$. For $0 < \kappa < \kappa_*$, choose $\lambda > 0$ with $D(P_\lambda, P) = \kappa$. Then

$$\mathbb{E}_{P_\lambda}[c] - \mu = \psi'(\lambda) = \frac{\kappa + \psi(\lambda)}{\lambda},$$

so this law attains the upper bound.

For $\kappa \geq \kappa_*$, the law $P(\cdot | \mathcal{A})$ is feasible and has cost m , the largest possible cost on the support of P . Also $(\kappa + \psi(\lambda))/\lambda \rightarrow m - \mu$ as $\lambda \rightarrow \infty$, proving equality in this case. For $\kappa = 0$, the only feasible law is P , and $\psi(\lambda)/\lambda \rightarrow 0$ as $\lambda \downarrow 0$. $\square$

For the absolute discrepancy, apply the proposition to both c and $-c$:

$$\sup_{Q: D(Q, P) \leq \kappa} d_c(P, Q) = \max_{s \in \{-1, 1\}} \inf_{\lambda > 0} \frac{\kappa + \psi(s\lambda)}{\lambda}.$$

This is the reverse-KL counterpart of the forward-KL modulus in Eq. (4.18). The subgaussian conversion replaces $\psi(s\lambda)$ by $\lambda^2 v_P(c)/2$. Any slack introduced at that step comes from summarizing the cost's full exponential-moment function by one quadratic bound.

4.B.4 A full-support version of the binary comparison

The infinite reverse KLs in the exact all-zero example are caused by its support. A small perturbation gives finite values and still distinguishes the generators. Take $P_\delta = \text{Bernoulli}(\delta)^{\otimes T}$; compare Q_{ind} from Exercise 5 with $Q_{\varepsilon, r}$ from Exercise 16. For $0 < \delta, \varepsilon, r < 1$, all three laws have full support. With $\text{kl}(a, b) = a \log(a/b) + (1-a) \log((1-a)/(1-b))$, the chain rule gives

$$\begin{aligned} D(P_\delta, Q_{\varepsilon, r}) &= \text{kl}(\delta, \varepsilon) + (T-1)[(1-\delta) \text{kl}(\delta, \varepsilon) + \delta \text{kl}(\delta, 1-r)], \\ D(Q_{\varepsilon, r}, P_\delta) &= \sum_{t=1}^T [(1-p_{t-1}) \text{kl}(\varepsilon, \delta) + p_{t-1} \text{kl}(1-r, \delta)]. \end{aligned}$$

Here p_t is the generated probability of a one, given by Eq. (4.91). For the independent law the two KLs are $T \text{kl}(\delta, \varepsilon)$ and $T \text{kl}(\varepsilon, \delta)$. At $T = 1000, \varepsilon = 10^{-3}, \delta = r = 10^{-6}$:

Q	$D(P_\delta, Q)$	$D(Q, P_\delta)$	$\mathbb{E}_Q[c_{\text{cnt}}]$
Q_{ind}	0.993	5.909	1
$Q_{\varepsilon, r}$	1.006	5082.198	368.224

The expected data cost is 0.001. Forward KL weights the transition out of one by the tiny data probability δ ; reverse KL weights it by the much larger generated probability p_{t-1} .

4.C The variance-sensitive bound and comparisons

4.C.1 Cost variance and Hellinger distance

The *Hellinger distance* between two laws P, Q is

$$h(P, Q)^2 = \sum_x (\sqrt{P(x)} - \sqrt{Q(x)})^2.$$

We use the normalization $0 \leq h(P, Q)^2 \leq 2$. Expanding the difference of square roots connects this distance to variation of the cost under P .

Proposition 4.C.1 (Hellinger distance and cost variance). *If $0 \leq c \leq C_{\max}$, then*

$$d_c(P, Q) \leq 2\sigma_P(c)h(P, Q) + C_{\max}h(P, Q)^2. \quad (4.94)$$

Moreover, $h(P, Q)^2 \leq D(P, Q)$.

Proof. Put $\Delta(x) = \sqrt{Q(x)} - \sqrt{P(x)}$ and $\bar{c} = c - \mathbb{E}_P[c]$. Expand

$$\mathbb{E}_Q[c] - \mathbb{E}_P[c] = 2 \sum_x \bar{c}(x) \sqrt{P(x)} \Delta(x) + \sum_x \bar{c}(x) \Delta(x)^2.$$

Cauchy–Schwarz bounds the first term in absolute value by $2\sigma_P(c)h(P, Q)$; $|\bar{c}| \leq C_{\max}$ bounds the second. For finite $\kappa = D(P, Q)$, Jensen’s inequality gives $\sum_x \sqrt{P(x)Q(x)} \geq e^{-\kappa/2}$, whence $h(P, Q)^2 \leq 2(1 - e^{-\kappa/2}) \leq \kappa$. $\square$

Substituting $h(P, Q)^2 \leq D(P, Q)$ into Eq. (4.94) proves the following restated result.

Proposition 4.2.9 (Cost variance and forward KL). *If $0 \leq c \leq C_{\max}$ and $D(P, Q) < \infty$, then*

$$d_c(P, Q) \leq 2\sigma_P(c)\sqrt{D(P, Q)} + C_{\max}D(P, Q). \quad (4.16)$$

Proof. Apply Eq. (4.94) and $h(P, Q)^2 \leq D(P, Q)$. Since $h(P, Q) \leq \sqrt{D(P, Q)}$, the stated bound follows. $\square$

4.C.2 Which bound is stronger? Two single-bit comparisons

One bit suffices to show that either the variance-sensitive or the subgaussian bound can be smaller. Throughout this subsection, $c(x) = x$ on $\{0, 1\}$, so $C_{\max} = 1$.

Proposition 4.C.2 (Comparison near a fair coin and near zero cost). *(a) Suppose*

$$P = \text{Bernoulli}(1/2), \quad Q = \text{Bernoulli}(1/2 + \delta), \quad 0 < |\delta| < 1/2.$$

As $\delta \rightarrow 0$, the Pinsker and subgaussian bounds are asymptotic to $d_c(P, Q) = |\delta|$, while the variance-sensitive bound is asymptotic to $\sqrt{2}|\delta|$.

(b) Suppose

$$P = \text{Bernoulli}(\varepsilon^2), \quad Q = \text{Bernoulli}(\varepsilon), \quad 0 < \varepsilon < 1.$$

As $\varepsilon \downarrow 0$, the actual discrepancy and the variance-sensitive bound are both asymptotic to ε . The Pinsker bound is asymptotic to $\sqrt{\varepsilon/2}$, and the subgaussian bound is asymptotic to $\sqrt{\varepsilon/\log(1/\varepsilon)}$.

Proof. For (a), direct expansion gives

$$\begin{aligned} D(P, Q) &= -\frac{1}{2} \log(1 - 4\delta^2) = 2\delta^2 + O(\delta^4), \\ D(Q, P) &= 2\delta^2 + O(\delta^4). \end{aligned}$$

Also $\sigma_P(c) = 1/2$ and $v_P(c) = 1/4$: every variance proxy is at least the variance, and Hoeffding's lemma gives equality for a fair coin. Thus the reverse-KL term of Eq. (4.15) is asymptotic to $|\delta|$. The minimum of its two terms is between the actual discrepancy and this term, so it has the same asymptotic value. Substitution into Eqs. (4.10) and (4.16) gives the other two comparisons.

For (b), the Bernoulli KL formula gives

$$\begin{aligned} d_c(P, Q) &= \varepsilon - \varepsilon^2 \sim \varepsilon, \\ D(P, Q) &\sim \varepsilon, \quad D(Q, P) \sim \varepsilon \log(1/\varepsilon), \\ \sigma_P(c) &= \varepsilon \sqrt{1 - \varepsilon^2} \sim \varepsilon. \end{aligned}$$

The variance-sensitive bound is therefore $2\sigma_P(c)\sqrt{D(P, Q)} + D(P, Q) \sim \varepsilon$. To evaluate the subgaussian bound, we use the sharp Bernoulli proxy [Schlemm, 2014, Eq. (1.1) and Theorem 1.1]:

$$v_{\text{Bernoulli}(p)}(c) = \frac{1 - 2p}{2 \log((1 - p)/p)}, \quad 0 < p < 1, \quad p \neq \frac{1}{2},$$

with value $1/4$ at $p = 1/2$. In particular,

$$v_Q(c) \sim \frac{1}{2 \log(1/\varepsilon)}, \quad v_P(c) \sim \frac{1}{4 \log(1/\varepsilon)}.$$

Consequently,

$$\begin{aligned} \sqrt{2v_Q(c)D(P, Q)} &\sim \sqrt{\frac{\varepsilon}{\log(1/\varepsilon)}}, \\ \sqrt{2v_P(c)D(Q, P)} &\sim \sqrt{\frac{\varepsilon}{2}}. \end{aligned}$$

The forward-KL term is the smaller one for sufficiently small ε . Pinsker gives $\sqrt{D(P, Q)/2} \sim \sqrt{\varepsilon/2}$. $\square$

Near a fair coin, Pinsker and the subgaussian inequality give the correct leading constant. Near zero cost, the variance-sensitive bound uses the small data variance to give the correct leading order, whereas even the optimal subgaussian proxies give a much larger bound. The first example requires square-root dependence on KL in a general cost bound. The zero-cost equality example in Section 4.2 requires linear dependence when the data variance vanishes. These explain the two terms in Eq. (4.16).

4.C.3 Moduli of continuity in analysis and approximation

A modulus of continuity quantifies how much a function's output can change when its input changes by at most a specified amount. For a function $h : [0, 1] \rightarrow \mathbb{R}$, its *modulus of continuity* is

$$\omega_h(\delta) = \sup_{\substack{x, y \in [0, 1] \\ |x - y| \leq \delta}} |h(x) - h(y)|.$$

Instead of only asking whether nearby inputs have nearby outputs, we ask how nearby the outputs must be. For example, a Lipschitz bound $|h(x) - h(y)| \leq L|x - y|$ gives $\omega_h(\delta) \leq L\delta$.

Proposition 4.C.3 (Quantitative uniform continuity). *A function $h : [0, 1] \rightarrow \mathbb{R}$ is uniformly continuous if and only if $\omega_h(\delta) \rightarrow 0$ as $\delta \downarrow 0$.*

Proof. Uniform continuity makes every difference in the defining supremum at most ε once δ is sufficiently small. Conversely, $\omega_h(\delta) < \varepsilon$ gives $|h(x) - h(y)| < \varepsilon$ whenever $|x - y| \leq \delta$. $\square$

In our comparison, the function is the expected-cost map $P \mapsto \mathbb{E}_P[c]$. We measure input discrepancy using KL and output discrepancy using d_c . Taking the supremum over P as well as Q gives the uniform comparison. If $\mathcal{P}$ contains all data laws, Pinsker's inequality gives $\omega_{D \rightarrow c}^{\mathcal{P}}(\kappa) \leq C_{\max} \sqrt{\kappa/2}$ for $0 \leq c \leq C_{\max}$: small KL uniformly forces nearby expected costs.

The expected-cost map can send distinct laws to the same value. Accordingly, d_c is a *pseudometric*: it is symmetric, satisfies the triangle inequality, and vanishes on equal laws, but can also vanish on distinct laws. The converse modulus quantifies how far apart the input laws can be while their expected costs remain close. Propositions 4.3.2 and 4.3.3 show how much this depends on the cost and reference law.

From continuity to approximation rates. A quantitative continuity bound can also tell us how accurately a function can be approximated by polynomials. Jackson's theorem states that, for a universal constant C , every continuous $h : [0, 1] \rightarrow \mathbb{R}$ satisfies

$$\inf_{\deg p \leq n} \|h - p\|_{\infty} \leq C \omega_h(1/n), \quad n \geq 1.$$

Thus $\omega_h(\delta) \leq L\delta^{\alpha}$, with $0 < \alpha \leq 1$, guarantees a degree- n approximation with uniform error at most $CLn^{-\alpha}$. The modulus turns a qualitative continuity statement into a bound on the degree needed for a given approximation error.

4.D Likelihood fitting and conditional support

4.D.1 Proof of the finite-class likelihood guarantee

Theorem 4.4.4 (Approximate maximum likelihood controls cost). *Let $\mathcal{Q}$ be a finite class containing P , and observe n independent sequences with law P . Let $0 \leq c \leq C_{\max}$. Suppose $\hat{Q} \in \mathcal{Q}$ satisfies Eq. (4.36) for $\eta \geq 0$. For $0 < \delta < 1$, put*

$$b = \frac{2 \log(|\mathcal{Q}|/\delta)}{n} + T\eta.$$

With probability at least $1 - \delta$,

$$d_c(P, \hat{Q}) \leq 2\sigma_P(c)\sqrt{b} + C_{\max}b. \quad (4.38)$$

In particular, if $\mathbb{E}_P[c] = 0$, then $\mathbb{E}_{\hat{Q}}[c] \leq C_{\max}b$.

Proof. The statistical step in [Theorem 4.4.4](#) controls Hellinger distance directly. For a fixed $Q \in \mathcal{Q}$, define its likelihood ratio on the observed sequences,

$$\Lambda_Q = \prod_{i=1}^n \frac{Q(X_{1:T}^{(i)})}{P(X_{1:T}^{(i)})}.$$

Independence of the sequences gives

$$\mathbb{E}[\sqrt{\Lambda_Q}] = \left(\sum_x \sqrt{P(x)Q(x)} \right)^n = \left(1 - \frac{h(P, Q)^2}{2} \right)^n \leq e^{-nh(P, Q)^2/2}.$$

Since $P \in \mathcal{Q}$, an η -approximate empirical minimizer satisfies

$$\widehat{L}(\widehat{Q}) \leq \widehat{L}(P) + \eta, \quad \Lambda_{\widehat{Q}} \geq e^{-nT\eta}.$$

For a fixed candidate Q , Markov's inequality therefore gives

$$\mathbb{P}(\Lambda_Q \geq e^{-nT\eta}) \leq \exp\left(\frac{nT\eta}{2} - \frac{nh(P, Q)^2}{2}\right).$$

If $h(P, Q)^2 > b = 2 \log(|\mathcal{Q}|/\delta)/n + T\eta$, this probability is less than $\delta/|\mathcal{Q}|$. A union bound over the class proves $h(P, \widehat{Q})^2 \leq b$ with probability at least $1 - \delta$. [Proposition 4.C.1](#) then supplies the cost bound. $\square$

The independent observations here are complete sequences. Dependence between tokens is part of their joint law P ; it never enters this likelihood-ratio calculation as an independence assumption. Lecture 6 studies how much information dependent observations supply.

4.D.2 Proofs of the parameter-sharing results

Two-sided risk and the leading constant.

Proposition 4.4.1 (Sequence KL under parameter sharing). *In the grouped independent-bit model, fit each probability by the pooled smoothed frequency from n independent length- T training sequences. Fix $0 < a < 1/2$ and suppose $p_j \in [a, 1 - a]$. Then*

$$\kappa_n(T) \asymp_a \frac{d(T)}{n}, \quad n, T \geq 1. \quad (4.31)$$

Here $\asymp_a$ means upper and lower bounds by positive constant multiples, with constants depending only on a . If $n \min_{1 \leq j \leq d(T)} m_j \rightarrow \infty$, then more precisely

$$\kappa_n(T) \sim \frac{d(T)}{2n}. \quad (4.32)$$

Proof of Proposition 4.4.1. First consider $B \sim \text{Binomial}(L, p)$, with $L \geq 1$ and $0 < p < 1$. The smoothed estimator is $\widehat{p} = (B + 1)/(L + 2)$. Using $1/(B + 1) = \int_0^1 u^B du$,

$$\mathbb{E}\left[\frac{1}{B + 1}\right] = \int_0^1 (1 - p + pu)^L du = \frac{1 - (1 - p)^{L+1}}{(L + 1)p}.$$

Consequently,

$$p \mathbb{E}[1/\widehat{p}] \leq \frac{L + 2}{L + 1}, \quad (1 - p) \mathbb{E}[1/(1 - \widehat{p})] \leq \frac{L + 2}{L + 1}.$$

The second inequality follows by applying the same calculation to $L - B$. Jensen's inequality gives

$$\begin{aligned}\mathbb{E}[\text{kl}(p, \hat{p})] &\leq p \log(p \mathbb{E}[1/\hat{p}]) + (1-p) \log((1-p) \mathbb{E}[1/(1-\hat{p})]) \\ &\leq \log \frac{L+2}{L+1} \leq \frac{1}{L+1}.\end{aligned}$$

For $p = 0, 1$, the loss equals $\log((L+2)/(L+1))$ deterministically, so the bound also holds.

For group j , the pooled count is binomial with $L = nm_j$. Independence of the test tokens makes the sequence KL the sum of the Bernoulli KLs. Therefore

$$\kappa_n(T) = \sum_{j=1}^{d(T)} m_j \mathbb{E}[\text{kl}(p_j, \hat{p}_j)] \leq \sum_{j=1}^{d(T)} \frac{m_j}{nm_j + 1} \leq \frac{d(T)}{n}.$$

For the lower bound, the binomial mean and variance give

$$\mathbb{E}[(\hat{p} - p)^2] = \frac{Lp(1-p) + (1-2p)^2}{(L+2)^2}.$$

Pinsker's inequality implies

$$\mathbb{E}[\text{kl}(p, \hat{p})] \geq 2\mathbb{E}[(\hat{p} - p)^2] \geq \frac{2Lp(1-p)}{(L+2)^2}.$$

With $L = nm_j$ and $p_j \in [a, 1-a]$,

$$m_j \mathbb{E}[\text{kl}(p_j, \hat{p}_j)] \geq \frac{2a(1-a)}{n} \left(\frac{nm_j}{nm_j + 2} \right)^2 \geq \frac{2a(1-a)}{9n}.$$

Summing over groups proves [Eq. \(4.31\)](#), with explicit bounds

$$\frac{2a(1-a)}{9} \frac{d(T)}{n} \leq \kappa_n(T) \leq \frac{d(T)}{n}.$$

For the leading constant, the same mean-squared error formula gives, uniformly for $p \in [a, 1-a]$,

$$\mathbb{E}[(\hat{p} - p)^2] = \frac{p(1-p)}{L} + O(L^{-2}).$$

In a fixed neighborhood of p , Taylor's theorem gives

$$\text{kl}(p, q) = \frac{(q-p)^2}{2p(1-p)} + O(|q-p|^3).$$

Binomial moment bounds give $\mathbb{E}[|\hat{p} - p|^3] = O(L^{-3/2})$. Outside that neighborhood the probability is exponentially small by a binomial tail bound, while $\text{kl}(p, \hat{p}) \leq \log(L+2)$. Splitting the expectation between the two events yields

$$\mathbb{E}[\text{kl}(p, \hat{p})] = \frac{1}{2L} + O(L^{-3/2}).$$

The constants can be chosen uniformly for $p \in [a, 1-a]$, $0 < a < 1/2$.

Summing over groups gives

$$\kappa_n(T) = \frac{d(T)}{2n} + O_a \left(\frac{1}{n^{3/2}} \sum_{j=1}^{d(T)} m_j^{-1/2} \right).$$

The relative error is $O_a((n \min_j m_j)^{-1/2})$, which tends to zero under the stated condition. This proves [Eq. \(4.32\)](#). $\square$

Learning rates and task-cost intervals.

Proposition 4.4.2 (Learning rates and task-cost intervals). *For any d with $1 \leq d(T) \leq T$, there exist a data law, terminal and counting costs, and estimator families such that, for all $n \geq 1, T \geq 2$, the estimators have common expected sequence KL $\kappa_n(T) \asymp d(T)/n$ and their expected costs range over exactly*

$$\left[0, 1 - e^{-\kappa_n(T)/2}\right] \quad \text{and} \quad \left[1 - e^{-\kappa_n(T)/2}, T(1 - e^{-\kappa_n(T)/2})\right],$$

respectively.

Proof. Let the U_t be independent fair bits and $V_t = 0$, giving one law P on infinite sequences. Use terminal cost V_T and counting cost $\sum_{t=1}^T V_t$. Since $1 \leq \lceil d(T) \rceil \leq T$, at each horizon we can partition the first T positions into $\lceil d(T) \rceil$ nonempty groups. The estimator may use a different partition at each horizon; the data law P remains fixed. Let $\hat{P}_T^U$ be the smoothed pooled estimator and set $r_T = \mathbb{E}[D(P_T^U, \hat{P}_T^U)]$. For every law R supported on $v_{1:T} \neq 0^T$, define

$$\hat{Q}_R = e^{-r_T}(\hat{P}_T^U \otimes \delta_{0^T}) + (1 - e^{-r_T})R.$$

Since all Bernoulli parameters are fixed at $1/2$, r_T is a deterministic function of the known sample sizes. Thus the prescribed mixture weight defines an estimator from the observed sample.

For this construction, $P_T = P_T^U \otimes \delta_{0^T}$. Since R assigns no mass to $v_{1:T} = 0^T$,

$$D(P_T, \hat{Q}_R) = D(P_T^U, \hat{P}_T^U) + r_T.$$

Hence every $\hat{Q}_R$ has expected sequence KL $\kappa_n(T) = 2r_T \asymp \lceil d(T) \rceil / n \asymp d(T)/n$. Both costs vanish under the first mixture component, so

$$\mathbb{E}[\mathbb{E}_{\hat{Q}_R}[c]] = (1 - e^{-r_T})\mathbb{E}_R[c].$$

On $v_{1:T} \neq 0^T$, terminal cost ranges from zero to one, while count cost ranges from one to T . Varying R over point masses and their mixtures gives the two intervals in [Proposition 4.4.2](#), since $r_T = \kappa_n(T)/2$. If an infinite-sequence output law is required, extend each fitted length- T law by a fixed continuation. $\square$

4.D.3 A sharp statistical obstruction to reverse-KL fitting

Suppose the learner must answer prompts drawn from a fixed law μ . Replaying a few training prompts would change the task: the distribution of prompts is supplied externally. This makes support recovery a conditional prediction problem.

Proposition 4.D.1 (Two candidate rules and rare distinguishing prompts). *Let $\mathcal{U}$ be finite, with $\mu(u) > 0$ for every u . Two known functions $f_0, f_1 : \mathcal{U} \rightarrow \{0, 1\}$ disagree on $\mathcal{A} \subseteq \mathcal{U}$, where $\mu(\mathcal{A}) = \alpha > 0$. Under candidate $i \in \{0, 1\}$, a demonstration has law*

$$P_i(u, y) = \mu(u)\mathbb{I}\{y = f_i(u)\}.$$

After n independent demonstrations, a possibly randomized learner returns a conditional law $\hat{q}$, inducing $\hat{Q}(u, y) = \mu(u)\hat{q}(y | u)$. Then

$$\inf_{\mathcal{L}} \max_{i \in \{0, 1\}} \mathbb{P}_{P_i^{\otimes n}, \mathcal{L}}(D(\hat{Q}, P_i) = +\infty) = \frac{1}{2}(1 - \alpha)^n. \quad (4.95)$$

Here $\mathcal{L}$ ranges over all such learning rules.

Proof. Finite reverse KL requires $\hat{q}(f_i(u) \mid u) = 1$ for every u ; hence it requires $\hat{Q} = P_i$. With probability $(1 - \alpha)^n$, none of the training prompts lies in $\mathcal{A}$. Conditional on this event the data have the same distribution under both candidates. No output can equal both P_0 and P_1 . Averaging over the two equally likely candidates, the probability of failure on this event is therefore at least $1/2$. This gives the lower bound on the worst-case failure probability.

For equality, use any observed prompt in $\mathcal{A}$ to identify i and return P_i . If none occurs, choose one of the two candidates with an independent fair coin. This has the stated failure probability under each candidate. $\square$

For $0 < \alpha < 1$ and $0 < \delta < 1/2$, success probability at least $1 - \delta$ requires

$$n \geq \frac{\log(1/(2\delta))}{-\log(1 - \alpha)}.$$

To make this concrete, a prompt is a pair of m -bit integers (L, R) , drawn uniformly and independently. The candidate rules are

$$f_0(L, R) = \mathbb{I}\{L \geq R\}, \quad f_1(L, R) = \mathbb{I}\{L > R\}.$$

They differ exactly when $L = R$, an event of probability 2^{-m} . The sample requirement is consequently $\Omega(2^m \log(1/\delta))$ for $0 < \delta \leq 1/4$. The class has only two rules; random demonstrations rarely distinguish them.

For comparison, let q give the common answer off $\mathcal{A}$ and a fair bit on $\mathcal{A}$. With $Q(u, y) = \mu(u)q(y \mid u)$ and task cost $c_i(u, y) = \mathbb{I}\{y \neq f_i(u)\}$, direct calculation gives

$$D(P_i, Q) = \alpha \log 2, \quad \mathbb{E}_{P_i}[c_i] = 0, \quad \mathbb{E}_Q[c_i] = \frac{\alpha}{2}, \quad D(Q, P_i) = +\infty.$$

Finite reverse KL demands exact conditional support even on very rare prompts. Forward KL and expected error can already be small before those prompts have been observed. One expert answer to a *chosen* prompt with $L = R$ identifies the rule. This specifies additional information that resolves the difficulty. Lectures 19 and 24 return to such differences between random examples and targeted queries.

4.E Variational objectives and the information they use

There are two distinct unknowns that may appear when evaluating a target probability: its normalization constant, and its relative probabilities at different sequences. A variational objective can eliminate the first from the optimization.

4.E.1 A target known up to normalization

Suppose a target law is specified as

$$P(x) = \frac{w(x)}{\mathcal{Z}}, \quad \mathcal{Z} = \sum_x w(x) > 0,$$

where w is evaluable but $\mathcal{Z}$ is expensive to compute. Then

$$D(Q_\theta, P) = \mathbb{E}_{Q_\theta}[\log Q_\theta - \log w] + \log \mathcal{Z}. \quad (4.96)$$

The final term is independent of θ . Minimizing reverse KL within a tractable family therefore uses samples from Q_θ and evaluations of $\log w$. This is a basic *variational approximation*: choose a tractable distribution by optimizing its divergence from a target.

For a differentiable model with fixed support contained in that of w , the gradient is

$$\mathbb{E} \left[(\log Q_\theta(\tilde{X}) - \log w(\tilde{X})) \nabla_\theta \log Q_\theta(\tilde{X}) \right], \quad \tilde{X} \sim Q_\theta.$$

Here $\tilde{X}$ denotes the whole generated sequence. Differentiating the expectation gives this identity; the additional normalization derivative vanishes because $\sum_x \nabla Q_\theta(x) = 0$.

When the target is an unknown data law available only through samples, an evaluable w has not been supplied. In particular, if $\hat{P}_n$ is the empirical law of training sequences and $\mathcal{S}_n$ is their finite set, then

$$D(Q, \hat{P}_n) < \infty \iff Q(\mathcal{S}_n) = 1.$$

Using a broader estimated density or a density-ratio estimator is an additional statistical modeling step.

4.E.2 A cost-defined target connects variational fitting to RL

Suppose instead that a cost $c(x)$ can be evaluated. Given a positive reference law R and $\beta > 0$, define

$$P_\beta(x) = \frac{R(x)e^{-\beta c(x)}}{\mathcal{Z}_\beta}, \quad \mathcal{Z}_\beta = \mathbb{E}_R[e^{-\beta c}].$$

Direct substitution gives

$$D(Q, P_\beta) = D(Q, R) + \beta \mathbb{E}_Q[c] + \log \mathcal{Z}_\beta. \quad (4.97)$$

Thus minimizing reverse KL to P_β is equivalent to minimizing

$$\mathbb{E}_Q[c] + \beta^{-1} D(Q, R).$$

The reference penalizes departures from an existing model, while the cost evaluates generated behavior. The supplied cost defines a new target whose relative probabilities can be evaluated. This is a useful connection to KL-regularized reinforcement learning and post-training.

4.E.3 Variational inference for the latent likelihood

For a hidden-state model, let $q_\theta(j, x)$ be the joint mass function of a hidden trajectory j and an observed sequence x . Let $r_\phi(j | x)$ be a tractable trial posterior whose support is contained in that of $q_\theta(j, x)$. Define

$$\mathcal{B}(\theta, \phi; x) = \sum_j r_\phi(j | x) \log \frac{q_\theta(j, x)}{r_\phi(j | x)}.$$

Using $q_\theta(j, x) = Q_\theta(x)q_\theta(j | x)$ gives

$$\log Q_\theta(x) = \mathcal{B}(\theta, \phi; x) + D(r_\phi(\cdot | x), q_\theta(\cdot | x)). \quad (4.98)$$

The *evidence lower bound*, or ELBO, is $\mathcal{B}$. Maximizing it fits the model and a tractable approximation to its hidden-state posterior. The divergence in Eq. (4.98) compares two laws of hidden trajectories conditional on a supplied observation. It measures the error in the likelihood approximation. The reverse sequence KL in Eq. (4.48) instead compares generated observation sequences with the data law.

4.F Proofs for the additional LDS results

4.F.1 Cayley–Hamilton

Theorem 4.7.2 (Cayley–Hamilton). *Let $A \in \mathbb{R}^{d \times d}$ have characteristic polynomial*

$$\chi_A(\lambda) = \det(\lambda I - A) = \lambda^d + a_{d-1}\lambda^{d-1} + \cdots + a_1\lambda + a_0.$$

Then

$$\chi_A(A) = A^d + a_{d-1}A^{d-1} + \cdots + a_1A + a_0I = 0. \quad (4.64)$$

Proof. The adjugate identity is

$$(\lambda I - A) \operatorname{adj}(\lambda I - A) = \chi_A(\lambda)I.$$

Write the adjugate, a matrix polynomial of degree at most $d - 1$, as $\sum_{j=0}^{d-1} \lambda^j D_j$. Comparing coefficients yields

$$D_{d-1} = I, \quad D_{j-1} - AD_j = a_j I \quad (1 \leq j \leq d-1), \quad -AD_0 = a_0 I.$$

Multiply the middle identities on the left by A^j , add them to the last identity and to $A^d D_{d-1} = A^d$. All the D_j terms cancel, leaving $\chi_A(A) = 0$. $\square$

4.F.2 The constant-gain Gaussian predictor

Theorem 4.7.4 (A constant-gain Gaussian predictor). *Consider Eq. (4.60) with spectral radius $\rho(A) < 1$, independent centered Gaussian process and observation noises with covariances $\Sigma_W \succ 0$ and $\Sigma_V \succ 0$. There are matrices $\Pi \succ 0$ and L such that, when $S_1 \sim \mathcal{N}(0, \Pi)$ independently of the noises, the recursion*

$$M_1 = 0, \quad M_{t+1} = FM_t + LZ_t, \quad F = A - LC \quad (4.66)$$

satisfies

$$M_t = \mathbb{E}[S_t | Z_{<t}], \quad Z_t | Z_{<t} \sim \mathcal{N}(CM_t, \Omega), \quad \Omega = C\Pi C^\top + \Sigma_V. \quad (4.67)$$

Moreover, $\rho(F) < 1$ and the observations have uniformly bounded second moments.

Proof. We construct the covariance used in [Theorem 4.7.4](#). If $S_t \mid Z_{<t} \sim \mathcal{N}(M_t, \Pi_t)$, Gaussian conditioning and the state equation give

$$\begin{aligned}\Omega_t &= C\Pi_t C^\top + \Sigma_V, & L_t &= A\Pi_t C^\top \Omega_t^{-1}, \\ M_{t+1} &= AM_t + L_t(Z_t - CM_t), \\ \Pi_{t+1} &= A\Pi_t A^\top + \Sigma_W - A\Pi_t C^\top \Omega_t^{-1} C\Pi_t A^\top.\end{aligned}\tag{4.99}$$

The covariance update is a deterministic map $\mathcal{T}(\Pi_t)$. To show that it has a fixed point under the stated conditions, use the identity

$$(A - GC)\Pi(A - GC)^\top + \Sigma_W + G\Sigma_V G^\top = \mathcal{T}(\Pi) + (G - L)(C\Pi C^\top + \Sigma_V)(G - L)^\top, \tag{4.100}$$

where $L = A\Pi C^\top (C\Pi C^\top + \Sigma_V)^{-1}$. For any vector, its quadratic form on the left is minimized over G by $G = L$. It follows that $\mathcal{T}$ is monotone in the positive-semidefinite order.

Since $\rho(A) < 1$, the finite matrix

$$B = \sum_{j=0}^{\infty} A^j \Sigma_W (A^\top)^j$$

satisfies $B = ABA^\top + \Sigma_W$. Starting from $\Pi_0 = 0$, iterate $\Pi_{n+1} = \mathcal{T}(\Pi_n)$. Monotonicity gives an increasing sequence, and $\mathcal{T}(B) \preceq ABA^\top + \Sigma_W = B$ bounds it above. It therefore converges to a fixed point Π . Continuity follows from $\Sigma_V \succ 0$, and $\Pi \succeq \Sigma_W \succ 0$.

Set $L = A\Pi C^\top (C\Pi C^\top + \Sigma_V)^{-1}$ and $F = A - LC$. The fixed-point identity becomes

$$\Pi = F\Pi F^\top + \Sigma_W + L\Sigma_V L^\top.$$

Thus $\Pi - F\Pi F^\top \succ 0$; conjugation by $\Pi^{-1/2}$ shows that $\rho(F) < 1$. Initialize $S_1 \sim \mathcal{N}(0, \Pi)$. Induction in [Eq. \(4.99\)](#) now keeps the error covariance equal to Π , proving [Eqs. \(4.66\)](#) and [\(4.67\)](#).

Finally,

$$S_t = A^{t-1} S_1 + \sum_{j=2}^t A^{t-j} W_j.$$

The geometric decay of powers of A bounds its second moment uniformly in t ; $Z_t = CS_t + V_t$ has the same property. Geometric decay of powers of F also supplies the constants G, γ in [Proposition 4.7.5](#). $\square$

4.F.3 Why the Hankel factors recover the predictor

Proposition 4.7.6 (Recovery of the prediction map). *The recovered matrices satisfy*

$$\overline{C} \overline{F}^{j-1} \overline{L} = K_j \quad \text{for every } j \geq 1.$$

Thus the recursion

$$\overline{M}_1 = 0, \quad \widehat{Z}_t = \overline{C} \overline{M}_t, \quad \overline{M}_{t+1} = \overline{F} \overline{M}_t + \overline{L} Z_t \tag{4.74}$$

computes the same predictor on every observed history.

Proof. Under the ranks assumed in [Section 4.7.3](#), both $\mathcal{H}_0 = \mathcal{O}_\star \mathcal{R}_\star$ and $\mathcal{H}_0 = \mathcal{O} \mathcal{R}$ are full-rank factorizations through a $d_\star$ -dimensional space. Their left factors have the same column space, so $\mathcal{O} = \mathcal{O}_\star T_0$ for an invertible T_0 . Multiplying by a left inverse of $\mathcal{O}_\star$ gives $\mathcal{R} = T_0^{-1} \mathcal{R}_\star$. Consequently, [Eq. \(4.73\)](#) yields

$$\overline{F} = T_0^{-1} F_\star T_0, \quad \overline{C} = C_\star T_0, \quad \overline{L} = T_0^{-1} L_\star.$$

Substituting these into $\overline{C} \overline{F}^{j-1} \overline{L}$ proves [Proposition 4.7.6](#). Small singular values make inversion sensitive to coefficient errors; the truncated SVD selects which directions to retain. $\square$

4.F.4 State-space error and long-run statistics

Consider two state processes on one probability space:

$$S_{t+1} = g(S_t) + W_{t+1}, \quad \tilde{S}_{t+1} = \hat{g}(\tilde{S}_t) + \tilde{W}_{t+1}.$$

Assume that $\hat{g}$ is ρ -Lipschitz, $0 \leq \rho < 1$, and that the local coupled RMS error is bounded:

$$\left(\mathbb{E} \left[\|g(S_t) + W_{t+1} - \hat{g}(S_t) - \tilde{W}_{t+1}\|_2^2 \right] \right)^{1/2} \leq \varepsilon.$$

With $e_t = (\mathbb{E}[\|S_t - \tilde{S}_t\|_2^2])^{1/2}$, inserting $\hat{g}(S_t) + \tilde{W}_{t+1}$ between the next states gives $e_{t+1} \leq \varepsilon + \rho e_t$. Thus [Eq. \(4.51\)](#) applies.

For observations $Z_t = r(S_t) + V_t$ and $\tilde{Z}_t = \hat{r}(\tilde{S}_t) + \tilde{V}_t$, assume that $\hat{r}$ is L_r -Lipschitz and

$$\left(\mathbb{E} \left[\|r(S_t) + V_t - \hat{r}(S_t) - \tilde{V}_t\|_2^2 \right] \right)^{1/2} \leq b.$$

The observation discrepancy satisfies

$$d_t := (\mathbb{E}[\|Z_t - \tilde{Z}_t\|_2^2])^{1/2} \leq L_r e_t + b.$$

For an L_f -Lipschitz function f ,

$$|\mathbb{E}[f(Z_t)] - \mathbb{E}[f(\tilde{Z}_t)]| \leq L_f d_t.$$

If both observation processes have RMS norm at most B_Z , then

$$\left\| \mathbb{E}[Z_t Z_s^\top] - \mathbb{E}[\tilde{Z}_t \tilde{Z}_s^\top] \right\|_F \leq B_Z (d_t + d_s). \quad (4.101)$$

To see this, expand the matrix difference using $(Z_t - \tilde{Z}_t)Z_s^\top + \tilde{Z}_t(Z_s - \tilde{Z}_s)^\top$, and apply Cauchy-Schwarz. Uniformly small coupled error therefore controls means, Lipschitz costs, and fixed-lag second moments.

For linear Gaussian models in common state coordinates, take $W_t = \Sigma_W^{1/2} \Xi_t$ and $\tilde{W}_t = \hat{\Sigma}_W^{1/2} \Xi_t$ with shared standard Gaussian Ξ_t , and couple the observation noises similarly. If the true states have RMS norm at most B_S , sufficient local bounds are

$$\begin{aligned} \varepsilon &\leq \|A - \hat{A}\| B_S + \|\Sigma_W^{1/2} - \hat{\Sigma}_W^{1/2}\|_F, \\ b &\leq \|C - \hat{C}\| B_S + \|\Sigma_V^{1/2} - \hat{\Sigma}_V^{1/2}\|_F. \end{aligned}$$

The contraction condition is $\|\hat{A}\| < 1$. A change of coordinates can express contraction in a weighted Euclidean norm. This is a concrete route from accurate, stable state-space fitting to long-run statistical accuracy.

Bibliographical notes

The probability and KL chain rules continue Lecture 3. The cost bounds and binary calculations here follow directly from those identities. The zero-cost forward-KL refinement makes explicit information that a general Pinsker conversion discards.

Likelihood, task discrepancy, and imitation learning. [Proposition 4.2.9](#) has Bernstein form, with a variance-dependent square-root term and a range-dependent linear term. [Foster et al. \[2024, Theorem 3.1\]](#) obtain this form using trajectory-level Hellinger distance and expert-action variance. Their lower-order term retains the total-return range R , up to logarithms. The technical gain in their Section 3.2 is to avoid an extra horizon factor: summing H advantages (changes in expected remaining return caused by an action) gives a naive range coefficient HR , which their concentration and stopping-time argument reduces to order R up to logarithms. Their Proposition 3.1 also bounds expert-action variance by the variance of the total return; a deterministic expert has zero action variance even in a random environment.

Here $\sigma_P(c)$ measures the variation of the specified trajectory cost directly. For our binary examples it is zero, and the sharp zero-cost bound already uses $C_{\max} = T$ for the count and $C_{\max} = 1$ for terminal error. [Wang et al. \[2025\]](#) survey variance-sensitive Hellinger bounds and the role of loss functions in decision making.

The finite-class likelihood proof in [Appendix 4.D.1](#) makes the optimization tolerance explicit. Lecture 18 develops the imitation-learning results, including the expert variance, parameter sharing, misspecification, and the sample-budget meaning of horizon dependence.

The comparison-rule example in [Appendix 4.D.3](#) adapts the rare-disagreement construction of [György et al. \[2025, Theorem 3.2\]](#) to conditional reverse KL. Its exact two-rule failure probability is proved here.

Minimax optimality of the parameter-sharing rate. The rate $d(T)/n$ in [Proposition 4.4.1](#) is optimal up to constants depending on a . Write P_p for the length- T grouped Bernoulli law with parameter vector p . Then

$$\inf_{\hat{Q}} \sup_{p \in [a, 1-a]^{d(T)}} \mathbb{E}[D(P_p, \hat{Q})] \asymp_a \frac{d(T)}{n},$$

where the infimum ranges over all learners that use n independent training sequences to output a probability law on length- T sequences. The upper bound follows from [Proposition 4.4.1](#). For the lower bound, put independent two-point priors on the parameters, with the two values near $1/2$ separated by a small multiple of $(nm_j)^{-1/2}$, depending only on a . At a test position in group j , the training data and preceding test tokens contain at most $(n+1)m_j \leq 2nm_j$ observations of that parameter. The two values remain hard to distinguish: Le Cam’s testing bound and Pinsker’s inequality give expected conditional KL at least a constant multiple of $1/(nm_j)$. Summing over the m_j positions per group and then over groups, using the KL chain rule, gives $d(T)/n$. [Yu \[1997\]](#) explains the two-point and coordinatewise testing methods underlying this argument.

Loss normalization in supervised fine-tuning. [Yao et al. \[2025, Section 4.3, Eq. \(6\)\]](#) distinguish averaging all predicted token losses from averaging separately normalized response losses, and compare these choices in supervised fine-tuning. [Hugging Face \[2024\]](#) explain how to implement token averaging across gradient-accumulation batches: sum the losses and divide by the total number of prediction targets included in the loss.

Statistical accuracy, optimization, and scaling laws. The strong-convexity certificate in [Section 4.4.3](#) follows by minimizing the quadratic lower bound in the definition of strong convexity; [Exercise 13](#) gives the calculation. [Boyd and Vandenberghe \[2004, Section 9.1.2\]](#) discuss this stopping criterion. [Bottou et al. \[2018, Theorem 4.7\]](#) give the $O(1/K)$ expected optimization rate used in [Endnote 4](#), under smoothness, strong convexity, controlled gradient noise, and suitable decreasing

stepsizes. Their Section 4.3 treats the weaker stationarity guarantees available for general nonconvex objectives.

[Bottou and Bousquet \[2007, Sections 2–3\]](#) study the approximation–estimation–optimization tradeoff. Balancing optimization tolerance against estimation error gives the statistical-accuracy principle used in [Eq. \(4.40\)](#). As data improve the statistical accuracy, finer optimization becomes worthwhile.

The connection to language-model scaling laws is explicit in [Hoffmann et al. \[2022, Section 3.3 and Appendix D.2\]](#). Renaming their variables, they fit the empirical relation

$$L(m, B) \approx L_\infty + a m^{-\alpha} + b B^{-\beta},$$

where m is parameter count, B is the number of training tokens, and the constants are fitted across training runs. Their last term combines finite-data effects and incomplete optimization during a single pass. Increasing the training-token budget supplies both observations and gradient updates. Under a fixed compute budget, [Kaplan et al. \[2020\]](#) find a regime in which a larger model stopped before convergence outperforms smaller models trained longer. This allocates computation between model size and fitting accuracy. These empirical laws describe average log loss. The cost comparisons in [Section 4.3](#) ask what task-performance variation remains at a given likelihood, and hence what additional information is useful for choosing a generator.

Stability and concentration. [Rudolf and Schweizer \[2018\]](#) develop perturbation bounds for Markov chains using contraction. [Theorem 4.5.2](#) is the elementary finite-state TV argument. [Raginsky and Sason \[2014, Sections 2.2.1 and 3.4.3\]](#) give Hoeffding’s lemma and the relative-entropy variational inequality. Our exponential-tilting calculation applies these ideas to a specified sequence cost. [Proposition 4.B.1](#) identifies its exact reverse-KL conversion by constructing extremizing tilted laws. The single-bit comparison in [Appendix 4.C.2](#) uses the sharp Bernoulli subgaussian proxy from [Schlemm \[2014\]](#).

Moduli of continuity and approximation. [Section 4.2.5](#) applies the usual quantitative continuity question to KL and expected cost. The fixed-KL endpoint calculations give both conversion directions directly. For the approximation-theory connection in [Appendix 4.C.3](#), [Drucker and de Wolf \[2011, Theorem 1\]](#) state Jackson’s theorem in the form used here and give proofs. Its modulus controls the uniform polynomial approximation error as a function of degree.

Variational inference. [Blei et al. \[2017, Section 2.2\]](#) derive the ELBO and its relation to reverse KL for posterior approximation. [Appendix 4.E](#) separates this computational use from fitting an unknown sequence-data law, and derives the cost-defined Gibbs target by direct substitution.

Linear prediction and system identification. The Kalman predictor is the classical Gaussian conditional-prediction recursion; [Mikusheva \[2009\]](#) give a derivation using Gaussian conditioning. The fixed-point and stability proof in [Appendix 4.F.2](#) supplies sufficient conditions for the version used here. Hankel factorization is the state-recovery construction of [Ho and Kalman \[1966\]](#). [Tsiamis and Pappas \[2019\]](#) study finite-sample stochastic system identification through observable regression and low-rank state recovery. Their analysis quantifies how data length and conditioning affect recovery; the exact algebra in [Section 4.7.3](#) explains the underlying construction.

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Endnotes

1. **Conventions at zero.** These conventions are the continuous extensions of the nonzero formula and express the infinite log loss incurred when a positive-probability outcome is assigned probability zero; Lecture 3, [Exercise 2\(e\)](#), verifies both limits.
2. **Variance proxy and variance.** How does the optimal proxy compare with the variance? Write $\sigma_P(c)^2 = \mathbb{E}_P[(c - \mathbb{E}_P[c])^2]$. Then

$$\sigma_P(c)^2 \leq v_P(c) \leq \frac{C_{\max}^2}{4} \quad \text{when } 0 \leq c \leq C_{\max}.$$

The lower bound follows by expanding [Eq. \(4.13\)](#) at $\lambda = 0$; the upper bound is Hoeffding’s lemma. Thus $\sqrt{v_P(c)}$, the subgaussian fluctuation scale, lies between $\sigma_P(c)$ and $C_{\max}/2$. For rare errors the proxy can greatly exceed the variance; [Appendix 4.C.2](#) gives a Bernoulli example. The bounds and existence of the minimum are proved in [Appendix 4.B.2](#).

3. **Unequal response lengths.** For response y_i to prompt u_i , let T_i count its predicted tokens and put $\ell_i(Q) = -\log Q(y_i | u_i)$. Two averaging conventions are

$$\hat{L}_{\text{token}}(Q) = \frac{\sum_i \ell_i(Q)}{\sum_i T_i}, \quad \hat{L}_{\text{response}}(Q) = \frac{1}{n} \sum_i \frac{\ell_i(Q)}{T_i}.$$

The first weights predicted tokens equally; the second weights each response’s average loss equally. Prompt and padding tokens excluded from the loss are also excluded from T_i . Both conventions reduce to the formula in [Section 4.4.3](#) when every $T_i = T$. Both are used in supervised fine-tuning; see the [bibliographical notes](#).

4. **Fixed passes and finer accuracy.** Consider smooth, strongly convex per-token objectives with constants uniform in n, T , bounded initial gaps, and unbiased stochastic gradients with uniformly bounded variance. With suitable decreasing stepsizes, stochastic gradient descent gives expected optimization error $O(1/K)$ after K updates. If batches contain a fixed number of uniformly sampled token losses, $K = \Theta(nT)$ updates process a fixed multiple of the dataset’s token count and give expected per-sequence error $O(1/n)$. This counts token-loss evaluations; their computational cost can depend on context length. The bound averages over optimizer randomness and does not certify an individual run. Full-gradient descent uses one pass per update, so fixed passes give fixed K . Even for $f(\theta) = \theta^2/2$, starting at $\theta_0 = 1$ with stepsize $1/2$ gives error 2^{-2K-1} , independent of T ; reaching s/T for fixed $s > 0$ requires order $\log T$ updates. Thus passes alone do not determine the achieved tolerance. The stochastic rate also need not hold for a general neural-network loss. Lecture 10 returns to when local gradients ensure global convergence; see the [bibliographical notes](#) for the assumptions and rate.
5. **Matrix and kernel conventions.** This convention matches column-vector state probabilities; the earlier kernel notation is $K(j, i) = A_{ij}$.